Techno-Economic Assessment of a Novel Multifeedstock Sustainable Aviation Fuel for Aviation Decarbonization

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Abstract— The mandate to decarbonize global aviation relies on the large scale deployment of drop-in Sustainable Aviation Fuels (SAF). However, the first-generation synthetic pathways is currently limited by single-feedstock dependency. This paper presents a techno-economic, regulatory, and risks assessment of a novel multi-feedstock SAF synthesized via an integrated hybrid biorefinery utilizing municipal solid waste (MSW), Jatropha, and livestock manure. To fully understand the reality of the feasibility of the project we will assess the reality of the certification state of the art and the economic aspect of it. Thermodynamic engine evaluations demonstrate that the fuel maintain safe combustion temperatures while delivering a 17% increase in engine thrust and a 10% extension in aircraft range compared to Jet A-1. Furthermore, a CFD study using COMSOL was conducted to evaluate the fuel hydraulic behaviour within a standard commercial wing pipeline. The simulations revealed minor localized pressure and mass flow drops due to fluid friction, which can be effectively mitigated through the cost-efficient installation of intermediate booster pumps. Economically and ecologically, this localized multi-feedstock architecture effectively neutralizes the risk of indirect land-use change and insulates aviation operators from global fossil fuel price volatility. This study concludes that the hybrid SAF model provides a highly competitive, scalable, and technically robust pathway for the immediate decarbonization of existing aviation fleets.

Keywords— SAF, Multi feedstocks, Cetrifications Standards, Risks, Techno-economic analyses, A320

I. Introduction: The Decarbonization Imperative and Techno-Economic Framework of Sustainable Aviation Fuels

Aviation is at a crossroads. Passenger demand keeps climbing, but so does the pressure to slash emissions. Before COVID, commercial flights pumped out roughly 915 million tonnes of CO_2 annually, 2.5% of total global CO_2, about 12% of transport-sector emissions (Yilmaz and Atmanli, 2017).

ICAO responded with aggressive targets. Under CORSIA, they want carbon-neutral growth and net-zero by 2050 (ICAO, 2022). At least 65% of those cuts have to come from drop-in Sustainable Aviation Fuel. Not hydrogen. Not batteries, but SAF. Why? Hydrogen combustion and battery-electric propulsion cannot handle heavy long-haul flights in the current state of the art (Chiaramonti, 2019). Jet A-1, the conventional fuel, comes from crude oil distillation. It is a hydrocarbon cocktail: paraffins, naphthenes, and 15–25% aromatics (Corporan et al., 2011). Those aromatics make the nitrile and elastomeric seals in fuel systems swell just enough to stop leaks at 35,000 feet.

First-generation SAF like HEFA and Fischer–Tropsch synthesis are nearly pure paraffinic, with almost zero aromatics (de Klerk, 2011). For 100% HEFA through an aircraft's fuel system, the seals shrink. Fluid failures follow. ASTM D7566 caps these synthetic fuels at 50% blending with Jet A-1 for exactly this reason (Rumizen, 2021).

But SAF has real advantages. No sulfur, no olefins. That basically means no SO_x emissions and reshapes soot formation pathways. Engine tests show that swapping to refined SAF can drop nvPM number emissions by 70% and cut contrail formation, addressing the non-CO_2 climate effects ICAO is concerned about (Schripp et al., 2022). Energy densities are better too. Synthetic paraffins pack 44.0–44.4 MJ/kg versus Jet A-1's 42.8, which translates to longer range or heavier payload on the same volume of fuel (Corporan et al., 2011).

We are business engineers of an airliner, the idea being to produce the most synthetic note to our superiors in order to help them decide on the relevance of focusing on a new SAF specifically. Our job is not to take a side, but instead to remain objective. That is where this hybrid SAF comes in (Alrebei et al., 2024). Engineered specifically to avoid first-generation problems, instead of relying on a single feedstock, it draws from municipal solid waste, *Jatropha curcas*, and livestock manure. HTL combined with steam gasification and Fischer–Tropsch upgrading creates an aviation hydrocarbon with 9.6 vol% aromatics (Alrebei et al., 2024), which falls within the ASTM D7566 limits of 8–25%. Moreover, it should be technically viable.

In the current geopolitical context, Sustainable Aviation Fuels (SAF) are not only a climate solution but also a strategic necessity. Ongoing conflicts and tensions have exposed the heavy dependence of the aviation sector on fossil-derived kerosene, whose supply is geographically concentrated and vulnerable to disruption. This creates significant risks in terms of price volatility, supply security, and political leverage. SAF offer a way to mitigate these risks by enabling more diversified and localized fuel production, based on widely available resources such as municipal waste, agricultural feedstocks, and biomass residues. By reducing reliance on imported oil, SAF enhance energy security and improve the resilience of the aviation sector against geopolitical shocks, making them a critical asset beyond their environmental benefits. The central question: does this multi-feedstock architecture actually deliver what ICAO needs to decarbonize global aviation, or is it just another dead end?

II. State of the Art

SAF and eSAF: Definition, Characteristics and Benefits

Sustainable Aviation Fuels (SAF) are alternative aviation fuels derived from renewable resources such as biomass, waste materials, or synthetic pathways, designed to reduce greenhouse gas (GHG) emissions over their entire life cycle. A key feature of SAF is their compatibility with existing aviation systems, as they can be used as "drop-in fuels", meaning they can be blended with conventional jet fuel without requiring significant modifications to aircraft engines or fuel infrastructure.

Among SAF, electrofuels (eSAF), also referred to as Power-to-Liquid fuels, are produced from low-carbon hydrogen and captured carbon dioxide. Hydrogen is typically generated through water electrolysis, while CO_2 is sourced from industrial emissions or direct air capture, and converted into liquid hydrocarbons via synthesis processes such as Fischer–Tropsch.

From a physical standpoint, SAF must exhibit properties similar to conventional Jet A or Jet A-1 fuels, including energy density, freezing point, viscosity, and thermal stability. These parameters are essential to ensure proper combustion, fuel system operation, and material compatibility within existing engines.

SAF offer significant environmental and operational benefits. They enable a reduction in life-cycle GHG emissions, depending on the feedstock and production pathway, and contribute to lower particulate emissions due to reduced sulfur and aromatic content. Their compatibility with existing aircraft allows for a progressive transition toward low-carbon aviation, while waste-based feedstocks support circular economy approaches.

However, several limitations remain. Feedstock availability is constrained and may compete with food production or lead to indirect land-use changes. In addition, production costs remain higher than those of conventional fuels, largely driven by feedstock prices and hydrogen requirements. For eSAF, the dependence on large quantities of low-carbon electricity represents a major challenge.

Performance and Market Development

Beyond environmental benefits, SAF can also influence aircraft performance. Studies conducted on turbofan engines, such as the CFM56-7B, have shown that certain SAF blends can improve combustion efficiency, leading to increases of up to 17% in thrust and 10% in range compared to conventional Jet A-1 fuel. These improvements are attributed to favorable thermodynamic properties similar to those of conventional fuels.

The development of SAF has evolved from experimental biofuels to certified aviation fuels integrated into commercial operations. Today, the sector is entering an industrialization phase supported by increasing investments and regulatory frameworks. Nevertheless, production volumes remain limited, and the market is still constrained by high costs and complex supply chains.

International Strategies and Regulatory Framework

The deployment of SAF varies across regions. The European Union relies on regulatory measures such as blending mandates and sustainability criteria to structure the market. In contrast, the United States focus on financial incentives and industrial scaling, while the United Kingdom combines regulatory objectives with industrial development strategies. In Asia, particularly in China, the availability of large feedstock resources offers strong potential, although further development depends on supply chain and policy improvements.

At the international level, the International Civil Aviation Organization (ICAO) coordinates efforts to reduce aviation emissions. The Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA) promotes the use of low-carbon fuels by defining sustainability criteria and emission accounting methods. In parallel, ASTM standards ensure fuel compatibility and define allowable blending limits for different SAF production pathways.

III. Evolution of SAF and Their Production Methods

Overview of SAF Production Pathways

Sustainable Aviation Fuel production relies on a range of biochemical and thermochemical pathways, which differ in feedstock requirements, technological maturity, and environmental performance. Initially based on simple lipid conversion processes, SAF technologies have progressively evolved toward more advanced and flexible systems, enabling improved yields and broader feedstock utilization.

Today, the sector is transitioning toward integrated biorefineries capable of processing multiple feedstocks. While some pathways have reached industrial maturity, others remain under development and are expected to contribute significantly to long-term aviation decarbonization.

Feedstocks and Resource Constraints

Feedstock selection plays a central role in determining SAF cost, scalability, and environmental impact. Current resources include waste oils, agricultural residues, lignocellulosic biomass, and, in the case of synthetic fuels, captured CO_2.

Waste-derived feedstocks, such as used cooking oil and animal fats, are economically attractive and compatible with

circular economy principles, but their limited availability restricts large-scale deployment. In contrast, lignocellulosic biomass offers greater potential but requires more complex conversion processes.

Synthetic pathways based on hydrogen and captured CO_2 provide an alternative with theoretically unlimited resources. However, their development is constrained by the high energy demand associated with hydrogen production and carbon capture technologies.

Main SAF Production Technologies

Several SAF production pathways have been certified or are under advanced development. The **HEFA** pathway is currently the most mature and widely deployed, relying on the hydroprocessing of lipid-based feedstocks. It produces a paraffinic fuel with favorable combustion properties and benefits from compatibility with existing refining infrastructure, although it remains limited by feedstock availability and hydrogen demand.

The **Fischer–Tropsch (FT)** pathway converts biomass or waste into liquid hydrocarbons via gasification and catalytic synthesis. It offers strong emission reduction potential and feedstock flexibility but requires complex and capital-intensive installations.

The **Alcohol-to-Jet (ATJ)** pathway transforms alcohols into jet fuel through multiple conversion steps, providing flexibility in feedstock selection at the expense of higher energy consumption and production costs.

The **SIP** pathway relies on fermentation processes to produce specific hydrocarbons, but its low yield and limited blending capability restrict its industrial relevance.

Finally, **Power-to-Liquid (PtL)** or eSAF pathways use hydrogen and captured CO_2 to produce synthetic fuels with near carbon-neutral potential. However, these technologies remain constrained by high costs, energy requirements, and limited infrastructure.

Comparative Analysis of SAF Pathways

A comparative analysis of the main SAF production pathways highlights trade-offs between maturity, cost, and environmental performance. As illustrated in **Table 1**, the HEFA pathway currently dominates due to its industrial maturity and relatively lower cost, making it the most deployable solution at present.

Fischer–Tropsch and Alcohol-to-Jet pathways represent intermediate solutions, offering greater flexibility and scalability potential despite higher complexity. In contrast, Power-to-Liquid pathways provide the highest decarbonization potential but remain limited by low maturity and high production costs linked to hydrogen and energy requirements.

These results indicate that no single pathway can meet future aviation demand alone, and that a combination of technologies adapted to available resources and regional conditions will be required.

Integrated Production Systems and Future Trends

Recent developments highlight the emergence of integrated and hybrid biorefineries capable of processing multiple feedstocks and producing both SAF and value-added co-products. This approach improves resource efficiency and economic viability by increasing system flexibility and reducing overall production costs.

Co-products such as biochar, bio-oil, or chemical intermediates contribute to revenue generation and help reduce the minimum fuel selling price. These integrated systems represent a key evolution in SAF production and are expected to play a central role in enabling large-scale deployment.

Pathway	Feedstock	Technological maturity	Cost level	Key advantage
HEFA	Oils and fats	High	Relatively low	Industrial deployment
FT	Biomass, waste	Medium–high	Medium	Strong GHG reduction
ATJ	Alcohols	Medium	Medium	Feedstock flexibility
SIP	Sugars	Low	High	Specific production route
PtL (eSAF)	H_2 + CO_2	Low	Very high	Carbon neutrality potential

TABLE 1: COMPARISON OF MAIN SAF PRODUCTION PATHWAYS

IV. NOVEL SAF STUDY

We focus our attention in this study on a novel Sustainable Aviation Fuel (SAF) synthesized via an integrated hybrid biorefinery model. Unlike conventional SAF that typically rely on a single dominant feedstock, this novel alternative uses diverse, multi-feedstock sources (Geleynse et al., 2018). By utilizing a combination of municipal solid waste (MSW), *Jatropha curcas* energy crops, and various livestock manures, this approach effectively avoids the supply chain limitations and land-use conflicts commonly associated with single-source biomass (Alherbawi et al., 2021).

This multi-feedstock strategy is directly served by an innovative amalgamation of several established production routes. The simulated architecture integrates HTL to process the wet biomass fraction into bio-oil without drying, alongside steam gasification to convert dry MSW into synthesis gas (Elliott, 2016). Then, FT synthesis and dry reforming are employed to polymerize the syngas into aviation hydrocarbons (de Klerk, 2011). The main advantages of this combined pathway is the natural generation of aromatic components. The HTL stream provides an aromatics content of 9.6%, which satisfies the ASTM D7566 used to prevent seal degradation, a common deficiency in pure FT-synthesized or Alcohol-to-Jet (ATJ) fuels (Rumizen, 2021).

The physicochemical and thermodynamic properties of this novel SAF present distinct advantages over other SAF. From a thermodynamic perspective, the fuel yields a comparable mass exergy of 1313.6 kJ/kg, producing the same power for the same input as jet fuel, and generates combustion flue gas temperatures of 1385°C, similar to the Jet A-1 (Alrebei et al., 2024). Consequently, it guarantees thermal compatibility with existing high-pressure turbine blades, eliminating the need for costly engine retrofitting. At the performance level, applying this SAF to a CFM56-7B turbofan demonstrates extraordinary gains, specifically a 17% increase in engine thrust and a 10% extension in aircraft range compared to conventional fuel (Alrebei et al., 2024). In addition, the integrated nature of the hybrid biorefinery presents notable economic advantages, estimating a cost reduction ranging from 0.13% to 8.08% compared to conventional fuels (Wang, 2016).

Despite these remarkable theoretical advantages, it is im-

portant to keep in mind that this novel SAF currently exists exclusively at the computational study level. Transitioning from simulation to a physical, industrial pilot plant remains a critical future prerequisite. The choice to investigate this specific novel SAF in this paper is justified by its potential to help aviation reach its decarbonization goal by offering a fully compatible, multi-source drop-in replacement that simultaneously enhances engine power output and range while mitigating feedstock issues; it represents a highly promising evolution. In the following part of our study, we will focus our attention on the properties of this fuel, and more importantly the viscosity of the blend as a function of its percentage, to analyse if the whole distribution system, both on the ground and in the aircraft, is well suited to this novel SAF. We will also assess its other techno-economic dimensions.

V. Certification Reality: Technical Qualification and Sustainability Recognition

Before this novel multi-feedstock SAF can be considered for commercial deployment, its certification pathway must be evaluated with caution. At this stage, the fuel is not yet a certified aviation fuel, but a promising candidate supported by hybrid biorefinery modelling, thermodynamic engine simulation, and preliminary system-level compatibility analysis (Alrebei et al., 2024). The study already provides a detailed performance assessment of the fuel on a CFM56-7B turbofan model, showing encouraging results in terms of combustion temperature, exergy behaviour, thrust and range (Alrebei et al., 2024). However, the final integrated multi-feedstock pathway itself still requires formal regulatory recognition (Rumizen, 2021).

The first certification barrier is technical qualification under ASTM D4054 (ASTM International, 2025a). This procedure defines how new aviation fuels and additives are evaluated before potential inclusion in ASTM D7566, the specification governing synthetic aviation turbine fuel components (ASTM International, 2025a; ASTM International, 2025b). The purpose is to demonstrate that the candidate fuel can be safely used in existing aircraft, engines, fuel systems and airport infrastructure without compromising safety, operability or durability (Rumizen, 2021; ASTM International, 2025a).

For a multi-feedstock SAF, this step is particularly critical. The use of municipal solid waste, Jatropha and livestock manure creates a flexible and resilient feedstock strategy, but it also introduces chemical variability (Alrebei et al., 2024; Alherbawi et al., 2021). The final fuel composition may change depending on the quality, seasonal availability and relative proportion of each input. Therefore, the fuel must be characterized through key physicochemical properties, including density, viscosity, freezing point, flash point, heating value, thermal stability, acidity, impurity content, aromatic fraction and overall hydrocarbon distribution (ASTM International, 2025a; ASTM International, 2025b).

Based on the available data, the studied SAF shows highly encouraging properties. Its density, viscosity, heating value, freezing point, flash point, sulfur content, olefin content and aromatic fraction are reported to be compatible with ASTM aviation fuel requirements (Alrebei et al., 2024; ASTM International, 2025b).

The ASTM D4054 process also includes fit-for-purpose evaluation. These tests assess combustion quality, storage stability, low-temperature behaviour, lubricity, deposit formation and compatibility with elastomers, seals, pumps, pipes and injectors (ASTM International, 2025a; Rumizen, 2021). In the reference study, the engine integration aspect has already been investigated through an Aspen Plus model of the CFM56-7B turbofan engine (Alrebei et al., 2024). The results show that the novel SAF produces combustion flue gas temperatures close to conventional fuels, with only a limited temperature difference compared with Jet A-1 (Alrebei et al., 2024).

The reported modelling results demonstrates technical feasibility and strong drop-in potential, but it does not by itself grant regulatory approval (Rumizen, 2021). The remaining certification challenge is therefore not to prove the concept from zero, but to translate these modelling results into a recognized qualification dossier under ASTM D4054, including the required fuel property documentation, component-level evidence, production consistency data and regulatory review (ASTM International, 2025a; CAAFI, 2026).

If the technical evaluation is accepted, the production pathway could be proposed for inclusion in ASTM D7566 (ASTM International, 2025b). Considering its paraffinic structure, low impurity content, thermal stability and aromatic content, the hybrid SAF can reasonably be compared to existing ASTM-recognized pathways such as HEFA-SPK, FT-SPK, ATJ-SPK or CHJ (ASTM International, 2025b; Rumizen, 2021). Under this hypothesis, it could potentially fall within the category of SAF components authorized up to 50% blending with conventional kerosene (ASTM International, 2025b). Nevertheless, this remains conditional on official ASTM D4054 qualification and ASTM D7566 annex recognition (ASTM International, 2025a; ASTM International, 2025b). Until this occurs, the fuel should be described as a high-potential drop-in candidate rather than a commercially certified drop-in fuel.

In parallel, technical certification is not sufficient to qualify the fuel as sustainable. A second certification layer is required to demonstrate environmental compliance. This evaluation is governed not by ASTM D4054, but by sustainability frameworks such as CORSIA, RED II / RED III, ReFuelEU Aviation, ISCC and RSB (ICAO, 2025a; European Union, 2023; European Commission, 2023; ISCC, 2026; RSB, 2020). These frameworks require proof of life-cycle greenhouse gas emission reduction, sustainable feedstock origin, traceability, land-use protection, biodiversity safeguards and social compliance (ICAO, 2025a; ICAO, 2025b; ISCC, 2026; RSB, 2020).

For the studied multi-feedstock SAF, sustainability certification is more complex than for single-feedstock pathways. Municipal solid waste, Jatropha and livestock manure each have different carbon intensities, transport constraints, land-use implications and regulatory treatment (Alrebei et al., 2024; ICAO, 2025a). The final environmental performance of the fuel therefore depends directly on the real contribution of each feedstock to the final product. A robust life-cycle assessment must therefore be conducted for each input stream and then weighted according to the actual fuel composition (ICAO, 2025a; ISCC, 2026).

Traceability is also a major certification requirement. The producer must document the origin, supplier, transport, conversion and incorporation of each feedstock (ICAO, 2025a; ICAO, 2025b; ISCC, 2026). This chain of custody is essential to ensure that non-eligible or high-impact materials are not hidden inside the multi-feedstock supply chain. In this sense, the flexibility of the hybrid model is both an advantage and a certification risk: it increases supply resilience, but also increases the burden of documentation and quality control (ICAO, 2025b; ISCC, 2026).

From a market perspective, certification must be designed internationally. In Europe, the fuel would need to comply with RED II / RED III and ReFuelEU Aviation (European Union, 2023; European Commission, 2023). In the United Kingdom, it would need to satisfy the UK SAF Mandate (UK Department for Transport, 2024). In the United States, eligibility would mainly depend on demonstrating sufficient life-cycle greenhouse gas reduction compared with conventional jet fuel (ICAO, 2025a). In Asian markets, recognition would likely rely on CORSIA-approved schemes such as ISCC CORSIA or RSB CORSIA (ICAO, 2025b; ISCC, 2026; RSB, 2020).

Overall, the certification outlook of this novel multi-feedstock SAF is promising but still unresolved (Alrebei et al., 2024; Rumizen, 2021).

Estimated certification cost

Since no public certification cost is available specifically for this multi-feedstock hybrid SAF, the following value must be treated as an order-of-magnitude estimate rather than a definitive project cost. Based on comparable ASTM D4054 qualification pathways, the technical certification budget could reasonably be estimated at 10–15 million USD, excluding the cost of producing the fuel volumes required for qualification (Gan et al., 2023; Heyne et al., 2021). This estimate is coherent with the staged ASTM process, where early fuel-property screening remains relatively limited in cost, while advanced component and engine-related certification stages dominate the total budget (ASTM International, 2025a; Heyne et al., 2021). In the present case, part of the technical risk is already reduced by the published Aspen Plus engine evaluation on the CFM56-7B, which demonstrated close combustion temperature, similar exergy behaviour, and improved thrust and range compared with Jet A-1 (Alrebei et al., 2024). However, this does not make the fuel formally certified under ASTM D7566 (ASTM International, 2025b). The remaining certification cost would mainly arise from the formal recognition of this hybrid production pathway, the verification of batch-to-batch fuel consistency, the documentation of impurities and material compatibility, and the regulatory review required for a new multi-feedstock route (ASTM International, 2025a; Rumizen, 2021). Sustainability certification would add additional but smaller costs, including life-cycle assessment, chain-of-custody documentation, ISCC or RSB audits, and annual certification fees (ICAO, 2025a; ISCC, 2026; RSB, 2020). Therefore, certification should be considered a significant financial risk, but the estimate remains uncertain because the exact cost can only be confirmed once the fuel enters an official ASTM qualification program (CAAFI, 2026; ASTM International, 2025a).

VI. Risk Assessment and possible Adaptation for the Use of a Novel Hybrid Feedstocks Sustainable Aviation Fuel

Introduction

The transition toward zero carbon in the context of the world's aeronautics goal relies heavily on the scalable deployment of Sustainable Aviation Fuel (SAF). The novel SAF under evaluation (synthesized via a highly integrated hybrid biorefinery) demonstrates true theoretical advantages, including a 17% increase in thrust and a 10% increase in aircraft range compared to conventional Jet A-1 fuel (Alrebei et al., 2024). This alternative drop-in fuel maintains thermodynamic behaviours nearly identical to conventional kerosene, theoretically minimizing the need for extensive engine modifications. Today, the American Society for Testing and Materials (ASTM) recognizes nine certified SAF production routes, which include HEFA, ATJ, and FT synthesis pathways (Alrebei et al., 2024), which are all part of the routes in the novel SAF. However, extrapolating these simulated successes to a commercial-scale production and distribution network introduces important feedstock, financial, technological, regulatory, and sustainability risks. Transitioning this novel fuel from a computed Aspen Plus simulation to a fully integrated industrial reality requires a deep evaluation of these risks and vulnerabilities.

Feedstocks Availability

The viability of any large-scale SAF initiative is fundamentally linked to the robustness of its feedstock supply chain. The proposed hybrid biorefinery process utilizes a highly diversified biomass portfolio including Jatropha crops, municipal solid waste (MSW), and varying livestock manures (wet sources). While this diversification theoretically dilutes the risk of single-source dependency, it introduces deep logistical and physicochemical complexities. Feedstock characteristics change a lot, for instance, MSW presents a moisture content of 7.56% and an ash content of 17.8%, whereas dairy manure exhibits an 85% moisture content and 32.7% ash (Alrebei et al., 2024). Handling such heterogeneous organic components necessitates sophisticated, energy-intensive preprocessing. The HTL process effectively mitigates some of these risks by continuously processing wet biomass into bio-oil without requiring thermodynamic drying (Elliott, 2016), while steam gasification converts the dry MSW into synthesis gas (CO and H_2) (Alrebei et al., 2024). The large-scale cultivation of energy crops like Jatropha risks initiating land-use conflicts (Alherbawi et al., 2021). If global demand scales disproportionately, it may exert pressure on arable land resources, sparking direct conflicts with the agricultural sector and exacerbating indirect land-use change (ILUC) dynamics, thereby threatening raw material supply stability. This competition problem can be present for all other feedstocks in the years to come, if, for example, they are used in other industries like transport, for example. We can currently see the great importance of not being dependent on international routes, and this point is a really great perk of this SAF.

Economic Risks

From a financial point of view, the commercialization of SAF has historically been marked by highly unfavorable economic parameters and the big risk of a low return on investment (ROI). Integrating multiple advanced conversion technologies (such as HTL, FT synthesis, and dry reforming) within a singular hybrid biorefinery demands massive capital expenditure (CAPEX) and elevated operational expenditure (OPEX) (Wang, 2016). Process simulations for comparable integrated biorefineries have calculated Minimum Jet-fuel Selling Prices (MJSP) with high variance as a function of the investment, the geographic zone, or even the feedstock availability (Kumar et al., 2018). Thus, the numbers are around an MJSP of 1.40 USD/kg by utilizing a 1.6 MT/year cane capacity, whereas they are at 1.69 USD/kg via alcohol-to-jet conversion. While the specific hybrid novel SAF studied estimates a cost reduction ranging from 0.13% to 8.08% compared to fossil-derived fuels (Alrebei et al., 2024), these figures remain deeply reliant on simulated mass-heat integrations and not on the true necessary volume of investment. Translating highly intensified thermal processes into full-scale industrial facilities introduces important financial risks due to scale-up inefficiencies, catalyst degradation over high-volume continuous cycling, unstable yield outputs, and potential changes in the distribution system (Geleynse et al., 2018). A failure to achieve precise economies of scale will invariably transfer the financial burden to commercial airlines, elevating the fare of passenger tickets and potentially rendering the biorefinery a stranded asset.

Technological Integration Challenge

Technological and integration risks take into account both the localized production phase of the novel SAF and its compatibility with the existing aircraft fleet. The drop-in concept demands that the synthesized fuel blends with petroleum-based jet fuel, satisfying all ASTM D7566 physicochemical mandates to avoid material deterioration. A critical historical risk of highly refined synthetic fuels is the complete absence of aromatic compounds, which are strictly required to prevent catastrophic fuel leakage (Yilmaz and Atmanli, 2017). The novel SAF successfully mitigates this specific integration risk; the integration of the HTL stream provides an aromatics content of 9.6%, comfortably satisfying the rigorous 8 to 25% asked by ASTM (Alrebei et al., 2024). At the engine level, thermodynamic evaluations utilizing the CFM56-7B turbofan model reveal that the combustion flue gas temperatures of the novel blend reach 1385 °C, exhibiting a negligible 5-degree variance compared to conventional jet fuels (1390.3 °C) (Alrebei et al., 2024). These precise metrics strongly suggest the fuel can be implemented without causing thermal degradation to high-pressure turbine blades. That said, it shows that this novel SAF theoretically lowers by a lot the risks of non-compatibility with the encountered systems. However, because this specific HTL-gasification and FT integration exists only in process simulation software, the technological production risk remains alarmingly high until real-size continuous flow, and it is the main technological risk of this novel SAF. Another important aspect is the distribution system and its compatibility with the existing infrastructure. The reality of this risk is assessed in this study.

Regulation Risks

Regulatory and certification landscapes present an additional number of risks, defined by rigid standards and geopolitical instability. To maintain certification as a commercial drop-in fuel, the novel SAF must strictly conform to ASTM D7566 limits regarding trace impurities (Rumizen, 2021), as well as the other certifications mentioned earlier. The simulation models clearly state that the SAF achieves a sulfur content of 1×10^{-4} % against an upper limit of 0.003 wt.%, and an olefins content of 0.87 % against an upper limit of 1.0 % (Alrebei et al., 2024). This model is not yet certified even if it uses certified routes and maintaining this highly precise chemical profile across fluctuating, seasonal biomass batches represents an important quality control risk in physical production. In addition, regulatory frameworks, such as the UN's CORSIA, or more importantly all national or regional regulations are strongly subject to political instability (Alrebei et al., 2024). Fluctuations in governmental subsidies, shifting carbon taxation models, or sudden alterations to the acceptable feedstocks mandated by the Renewable Energy Directive (RED II) could instantly impact and change the regulatory or even the economic viability of the fuel. Producers face the persistent risk that a sudden regulatory shift could decertify their specific biomass supply chain, forcing cost-prohibitive compliance reporting and halting commercial distribution.

Durability Risks

The last kind of risk that can threaten the SAF is durability and lifecycle risks. The intensive industrial unit operations required to run a hybrid biorefinery (including high-pressure steam gasification, catalytic hydrotreating, and dry reforming of methane and CO_2) possess immense baseline energy and utility demands (Hannula and Arpiainen, 2015). There is a pronounced durability risk that the supply chain, when accounting for agricultural harvesting, wet biomass transport, and intensive thermochemical processing, could emit excessive quantities of CO_2. If the production relies on carbon-intensive energy, these emissions will negatively impact the carbon reductions achieved during the aircraft's cruising phase. Even if the simulated biorefinery integrates on-site power generation via a combustion-turbine-generator and methanol-based carbon capture units to achieve energy self-sufficiency, comprehensive lifecycle assessments (LCA) are critically necessary to empirically verify the well-to-wake carbon intensity of this fuel (Martinez-Hernandez et al., 2019). In addition, to prevent these risks, important tracking of the origin of the SAF must be created to ensure trust in the sources.

So, while the deployment of this novel SAF contains highly complex risks (risk matrix Fig. X), its distinct hybrid architectural design inherently lowers some critical vulnerabilities that historically impact alternative aviation fuels. By synthesizing diverse solid and wet feedstocks via integrated HTL and gasification, the framework actively dilutes the supply chain risks associated with single-source crop failures. The fuel's properties alignment with conventional Jet A-1 effectively lowers the most prohibitive technological risk: the necessity for multi-billion-dollar hardware modifications to the existing commercial CFM56-7B aircraft fleet. Despite this fact, transitioning from Aspen Plus steady-state modeling to continuous commercial production requires extensive empirical validation

to fully de-risk this promising decarbonization strategy.(cf. Table 2)

Gravité ↓ Probabilité →	Low	Medium	High
Critical	Compatibility technologic (Drop-in)	Feedstock logistic	Industrial scale
Major	Certification	Land Use Conflict	ROI
Moderate	|	Life Cycle Emission	Reglementation instability
Mineure	|	|	|

TABLE 2: RISK MATRIX

VII. TECHNICAL STUDY

Framework of the technical study: fluid dynamics and fleet integration

To ground the techno-economic analysis of the Novel SAF in engineering reality, this study focuses on the critical phase of fleet integration: fuel delivery from wing tanks to the engine. Although certified SAFs must meet international ASTM standards, minor variations in density and viscosity can significantly alter fluid friction and lead to pressure losses within complex aircraft piping. These losses directly impact the stability and volume of fuel reaching high-pressure pumps, posing operational and safety risks. Due to computational and mesh size constraints, the simulation analyses a localized 2-meter pipe section to precisely measure these hydrodynamic behaviours within the existing architecture of an Airbus A320.

To add more context to this scale, we must look at the Airbus A320, which is used as the base for this study due to the fact that he is the most common commercial aircraft in the world. An Airbus A320 has a wingspan of roughly 34 meters. The internal fuel system, which draws from the centre tank, inner wing tank, and outer wing tanks, relies on a primary gallery of tubing to deliver fuel to the engine pylons. The main fuel feed line spanning from the inner wing pumps to the engine inlet run approximately 15 to 20 meters in total length per wing. From that we deduce that the 2-meter section simulated in our study represents 10% to 13% of the total primary fuel alimentation pipe length for a standard A320. The primary goal of this targeted study is to quantitatively measure the hydrodynamic behaviour of the novel SAF, to determine the possibility of its rapid, unmodified deployment, and determine exactly what mechanical hardware modifications are necessary to guarantee safe engine supply.

The study: simulation parameters and fluid properties

To quantify the fluid behaviour of this novel SAF against traditional benchmarks, we refer to the study we ran utilizing COMSOL Multiphysics software to model the fluid dynamics of the flow. To ensure a comprehensive analysis that shows well the realistic, phased integration of SAF into commercial aviation, we established four distinct test scenarios. These scenarios correspond to different levels of SAF integration: 100% pure Jet-A1 kerosene (the current global baseline), 95% Kerosene / 5% SAF blend (representing early-stage drop-in integration), 50% Kerosene / 50% SAF blend (representing the maximum allowable blending limit under current ASTM D7566 regulations), and 100% pure SAF (the ultimate future goal for total decarbonization). The parameters governing the simulation in the study we ran are rooted in the precise mass and kinematic properties of the fuels. (cf. Table 3)

	density ρ	kinematic viscosity ν	dynamic viscosity μ = ρ × ν
Jet-A1 kerosene	800 kg/m^3	5 × 10^6 m^2/s	4.0 × 10^3 Pa·s
SAF	791.4 kg/m^3	3.95 × 10^6 m^2/s	3.13 × 10^3 Pa·s

TABLE 3: PURE FUELS PROPERTIES

For the intermediate blends, the mass properties were calculated proportionally, with the 95/5 blend and 50/50. (cf. Table 4)

	density ρ	kinematic viscosity ν	dynamic viscosity μ = ρ × ν
95% Jet-A1 kerosene / 5% SAF	799.57 kg/m^3	4.9475 × 10^6 m^2/s	3.96 × 10^3 Pa·s
50% Jet-A1 kerosene / 50% SAF	795.7 kg/m^3	4.475 × 10^6 m^2/s	3.56 × 10^3 Pa·s

TABLE 4: INTERMEDIATE BLENDS PROPERTIES

These fluid parameters were set into the 2-meter, 3D pipe geometry to observe standard flow characteristics. The exhaustive numerical outputs of these four distinct CFD tests are compiled and can be directly referenced in the results table under the (cf. Table 5). It is important to remember the computational limits of our student licenses during this phase. The academic licensing puts limits on the maximum number of structural nodes and degrees of freedom within the finite element analysis. Thus, it has prevented us from simulating the entire 20-meter wing gallery with all its complex valves and cross-feed junctions simultaneously. In addition, we must state that the numerical value derived from these simulations are fundamentally based on fluid theory and steady-state modelling. However, as long as we are not able to conduct real-life flow tests on this newly synthesized SAF using actual Airbus hardware, this mathematically rigorous simulation remains the most precise and reliable predictive model we can utilize to guide engineering decisions.

While the Novel SAF is chemically compatible, it is hydraulically less “robust” than kerosene. This creates a pressure deficit that could lead to engine fuel starvation if not addressed. We came to this conclusion based on an additional analysis of the results in Appendix 2.

Results and consequences: hydrodynamic impacts and hardware solutions

The comparative analysis extracted from the study we ran shows a consistent, progressive shift in the flow parameters as the volumetric proportion of SAF increases in the fuel mixture. By analyzing the data from (cf. Table 5), we observe some critical hydrodynamic consequences when transitioning from standard Jet-A1 to 100% pure SAF. First, there is a distinct decrease in the mass flow rate, which drops from a robust 12.636 kg/s for pure kerosene down to 11.781 kg/s for the pure SAF. Second, we note a slight but measurable decrease in the volumetric flow rate falling from 0.015795 m^3/s to 0.014889 m^3/s. Third, the simulation highlights a reduction in

the average exit velocity, which decelerates from 22.933 m/s to 21.487 m/s. The total pressure measured at the outlet drops significantly from 3696.3 Pa to 3274.6 Pa.

Fuel	Pressure (Pa)	Mass flow rate (kg/s)	Volumetric flow rate (m^3/s)	Velocity (m/s)
Pure kerosene (Jet-A1)	3696.3	12.636	0.015795	22.933
95% kerosene / 5% SAF blend	3608.5	12.482	0.015612	22.641
50% kerosene / 50% SAF blend	3437.8	12.094	0.015241	22.084
Pure SAF	3274.6	11.781	0.014889	21.487

TABLE 5: COMPARAISON OF THE 4 STUDIED FUEL TYPES

These results clearly translate to a less favourable hydraulic behaviour for the SAF within the geometry of the wing. This behaviour is entirely coherent with the fluid dynamics of a bent of pipe. In piping networks, elbows and sharp bends creates minor pressure losses due to flow separation and secondary turbulent vortexes. Because the pure SAF has a lower density and a different dynamic viscosity profile than standard Jet-A1, the fluid's ability to maintain its momentum through the structural bends of the wing pipe is compromised, resulting in a higher pressure drops. If this unmitigated pressure loss were scaled across the entire 20-meter fuel line of an A320, it could disrupt the required baseline pressure at the engine inlet, leading to fuel starvation or engine cavitation during high-demand transient flight phases, such as take-off.

To solve this deficit, a pragmatic technical solution must be implemented: the addition of intermediate fuel booster pumps in the aircraft's fuel gallery. By artificially restoring the pressure head lost to the fluid friction and elbow bends, the mass and volumetric flow rates can be recovered to match the demand of the turbofan engines. To be stay based on the scale of an A320, the modification is manageable. An A320 relies on two main pumps in the centre tank and two main pumps in each inner wing tank. To compensate the pressure losses identified in our study, engineers would only need to install one to two additional, compact inline booster pumps per wing line (totalling two to four additional pumps for the entire aircraft). The structural feasibility of this modification is high. Modern commercial wings contain adequate maintenance access panels and internal bay space to accommodate compact inline pumps, and the existing electrical system of the A320 can easily support the minor amperage draw required by these supplementary units.

VIII. ECONOMIC VIABILITY OF THE MULTI-FEEDSTOCK HYBRID BIOREFINERY FOR SUSTAINABLE AVIATION FROM OUR PERSPECTIVE

The financial viability of retrofitting

When advancing technical modifications for commercial aviation, establishing a direct link with the economic realities of fleet management is necessary for our techno economic analysis. The proposition of adding structural hardware to an aircraft is often confronted to financial problems, but a comparative economic analysis justifies this specific adaptation.

Based on current aviation supply chain figures, a standard, certified electric aircraft fuel booster pump (similar to those manufactured for narrow-body commercial jets) carries an acquisition cost of $30,000 and $50,000 depending on the specifications. If an airline operator needs to add a maximum of four intermediate pumps to an A320 to perfectly optimize the delivery of 100% pure SAF, the total hardware cost would equal approximately $120,000 to $200,000 per aircraft. The labor and certification costs associated with this retrofitting are also easily mitigated, as these pumps can be seamlessly integrated during the aircraft's scheduled "C-check" or "D-check" heavy maintenance windows, meaning the aircraft does not suffer from unexpected out-of-service downtime.

To understand the true economic viability of this technical solution, we must compare the cost of adding these pumps to the capital required for a complete fleet overhaul. As of the current market in 2024–2025, the list price of a single, brand-new Airbus A320neo sits at roughly $88 million to $100 million. The cost of integrating the necessary booster pumps ($200,000) represents a nearly negligible 0.2% of the cost of purchasing a newly designed aircraft built specifically for alternative fuels. Therefore, the results of the study we ran conclusively demonstrate that while the novel SAF introduces minor hydrodynamic challenges regarding pressure drops, the solution is astonishingly cost-effective. It proves that the global commercial aviation sector can easily and affordably adapt its existing, legacy fleet to operate on 100% sustainable fuels in the immediate future, avoiding the crippling financial burden of replacing thousands of functional airframes. However we propose a complementary economic approach to this by seeing the economic viability of the production route.

Production costs: CAPEX, OPEX and the weight of feedstock

Integrating multiple advanced conversion technologies within a singular hybrid biorefinery demands massive capital expenditure (CAPEX) and elevated operational expenditure (OPEX) (Wang, 2016). For comparable integrated biorefineries, process simulations have calculated Minimum Jet-fuel Selling Prices (MJSP) with high variance depending on investment scale, geographic zone, and feedstock availability (Kumar et al., 2018). Concrete figures from the literature put the MJSP around 1.40 USD/kg when utilizing a 1.6 MT/year sugarcane capacity, and at 1.69 USD/kg via alcohol-to-jet conversion (Kumar et al., 2018). At the scale of 60 million gallons per year using FT technology, Pavlenko et al. (2019) calculated MJSP values of $1.53/L with municipal solid waste, $1.98/L with agricultural residue, and $2.15/L with energy crops. In most SAF routes, the feedstock price alone accounts for 60% to 75% of the total fuel cost (Alherbawi et al., 2021). Operating costs represent the largest share of total expenditure, driven by the maintenance of reactors, separation facilities, and purification systems (Wang et al., 2024). Additionally, the HEFA and FT-SPK routes generate by-products such as glycerol and solid waxes that require extra separation steps, further inflating OPEX (Wang et al., 2024).

The hybrid biorefinery addresses this partly through feed-

stock diversification. By processing Jatropha crops, municipal solid waste (MSW), and livestock manures simultaneously, the facility avoids single-source dependency (Alrebei et al., 2024). MSW has very low or even negative acquisition costs in many municipalities, while Jatropha thrives on marginal, non-arable land without competing with food production (Alherbawi et al., 2021). Internal heat and mass integration between process streams reduces utility demands, and on-site power generation via a combustion-turbine-generator approaches energy self-sufficiency (Alrebei et al., 2024). For comparison, the HEFA route, currently the most mature commercially, achieves an MJSP of $2.22 per kg with a capital investment of about $179.4 million (Diederichs et al., 2016). The hybrid targets a comparable cost range while producing a fuel that additionally increases thrust by 17% and range by 10% (Alrebei et al., 2024), gains that translate directly into lower fuel burn per passenger-kilometer.

Infrastructure, storage, and distribution: the drop-in advantage

One of the strongest economic arguments for this SAF is its drop-in character. A drop-in fuel is functionally equivalent to petroleum fuels and fully compatible with existing petroleum infrastructure (Shahriar and Khanal, 2022). The chemical composition of the novel blend, close to fossil kerosene, allows it to be incorporated without degrading engine performance, without threatening flight safety, and without requiring modification of transport, storage, and distribution networks already in place (IFPEN, 2023). Building new pipelines, airport hydrant systems, or tank farms would add hundreds of millions in capital costs and years of regulatory delay; none of that is needed here. The SAF volumetric energy density stands at 34.1 MJ/L versus 35.06 MJ/L for Jet A-1 (Lau et al., 2024), ensuring identical handling characteristics throughout the existing supply chain. Unnecessary costs such as altering or updating engines, transportation, and storage systems can be avoided by keeping fuel qualities constant (Lau et al., 2024). Storage costs therefore remain unchanged: the same tanks, the same pumps, the same quality control procedures apply. For airlines, this means zero capital outlay on ground infrastructure upgrades, a fact that makes SAF considerably more attractive than hydrogen, which demands entirely new cryogenic systems and incurs a 25% increase in aircraft production and maintenance expenses (Lau et al., 2024).

Local production and price stability

The most strategically significant argument for the hybrid biorefinery is that it enables local, decentralized fuel production. Today, conventional jet fuel prices follow global crude oil markets. The U.S. Energy Information Administration reported a 2.7% expected annual growth in jet fuel price from 2016 to 2050 (Shahriar and Khanal, 2022). Geopolitical instability, from Middle Eastern conflicts to the Russia-Ukraine war, translates directly into unpredictable kerosene price spikes. A significant shift to SAF can mitigate the risks with disruptions in the supply chain of jet fuel due to price volatility and geopolitical situations (Shahriar and Khanal, 2022). The hybrid biorefinery, by sourcing feedstocks locally – municipal waste from the surrounding region, Jatropha from nearby marginal lands, manure from local farms – effectively decouples aviation fuel cost from global petroleum dynamics. The supply chain is shortened from intercontinental to regional, and price signals become a function of local biomass availability rather than OPEC or sanctions regimes. A mature SAF market will foster energy security by reducing dependency on fossil fuels, thus enhancing geopolitical stability, and the diversification of energy sources will protect aviation from the volatility of oil prices and supply disruptions (Wang et al., 2024). For airline finance departments, the capacity to lock in long-term feedstock contracts with local suppliers represents a hedging mechanism that no petroleum futures market can replicate.

Maintaining affordable ticket prices

Jet fuel accounts for roughly one-third of an airline's operational expenditure (Alherbawi et al., 2021), so even a small premium can ripple through to the final ticket. Several mechanisms work together to contain this green premium. First, blending dilutes the impact: since SAF is currently mixed at ratios up to 50% with conventional kerosene (Shahriar and Khanal, 2022), the price differential is halved before reaching the fuel bill. Calculations on ATJ-derived triisobutane showed that the additional cost for a 1,000 km intra-EU flight ranged from only €1.20 to €4.30 per passenger (Lau et al., 2024), roughly 1 to 3% of a typical round-trip ticket. Second, governments provide incentives specifically designed to close the gap. The US Sustainable Aviation Fuel Act established a tax credit of $1.50 to $1.75 per gallon depending on the GHG reduction achieved (Shahriar and Khanal, 2022). ReFuelEU mandates 2% SAF in 2025, scaling to 63% by 2050 (Shahriar and Khanal, 2022), creating the demand certainty needed for scale-up. CORSIA further incentivizes adoption by reducing carbon offset obligations (Shahriar and Khanal, 2022). Third, economies of scale are especially strong: doubling HEFA plant capacity with jatropha oil cut MJSP from $3.82/gallon to $3.51/gallon (Shahriar and Khanal, 2022), and scaling ATJ production from 20 to 100 million gallons per year reduced it from $1.79/L to $1.14/L (Shahriar and Khanal, 2022). Fourth, co-product revenues help considerably: Klein et al. achieved a negative MJSP of -$0.83/gallon by selling hydrous ethanol and green naphtha from an FT-sugarcane biorefinery (Shahriar and Khanal, 2022). The hybrid model generates similar valorizable streams. Finally, the 17% thrust increase and 10% range gain (Alrebei et al., 2024) mean less fuel burned per seat-kilometer, which partially offsets any residual premium at the pump.

IX. Aviation Fuel Key Opportunities

Getting the aviation sector fully decarbonized means moving beyond theoretical fuel synthesis and into real, commercially viable deployment. The hybrid Sustainable Aviation Fuel (SAF) we tested shows remarkable benefits with a 17% increase in engine thrust and 10% better range, but those numbers only matter if we can navigate the realities of aircraft modifications, environmental life-cycle impacts, and economics (Alrebei et al., 2024). Airlines cannot afford to go bankrupt chasing green credentials, and we cannot swap one environmental problem for another. This section breaks down what makes this SAF ecologically different, what hardware changes legacy fleets need, how to keep ticket prices

reasonable, and how we can actually scale production.

Why This Fuel Is Ecologically Different

Most contemporary SAF production (especially HEFA or conventional ATJ pathways) relies heavily on single agricultural feedstocks like soy or lipid-producing sugarcane (Kumar et al., 2018). This creates a land-use problem. When you expand energy crops aggressively, you displace food agriculture, drive deforestation, and release massive amounts of carbon stored in soil. The emissions you "save" in the air get canceled out by what you release on the ground, a phenomenon called Indirect Land-Use Change (ILUC) (Alherbawi et al., 2021).

Our hybrid SAF sidesteps this entirely. Instead of competing for farmland, it pulls from municipal solid waste (MSW), agricultural residues, and livestock manure through Hydrothermal Liquefaction (HTL) and steam gasification (Elliott, 2016). This does two things. First, it decouples fuel production from arable land, so there is zero threat to food security. Second, it acts as waste management. Using MSW and wet manure prevents them from rotting in landfills, which would otherwise release methane, a greenhouse gas far more potent than CO_2. It shows the difference between our SAF and other SAF.

The fuel itself is cleaner too. With 9.6 vol.% aromatics and sulfur at just 1×10^{-4} % (Alrebei et al., 2024), it drastically cuts non-volatile particulate matter (nvPM) and sulfur oxide (SO_x) emissions during cruise. This is not just carbon-neutral, it is actively restorative; it differentiates it from other SAF fuel.

Hardware Changes for Existing Fleets

The new SAF burns a lot like conventional Jet A-1; combustion temperatures and mass exergy are nearly identical, so high-pressure turbine blades will not need redesigning. But there is one mechanical consideration. Minor differences in kinematic viscosity mean the fuel interacts slightly differently with the internal surface roughness of legacy fuel lines. This can increase fluidic friction, leading to localized pressure drops between the main tanks and the engine's high-pressure injection system (Moir and Seabridge, 2008).

The fix is straightforward: install an intermediate booster pump in the fuel delivery system. Yes, this adds weight. In aviation, every kilogram matters, so it is very important. But here is the thing: the performance gains from the fuel itself, 17% more thrust, 10% better range, more than compensate for the pump's weight penalty. And installing a booster pump is not very complex engineering. It is a standard retrofit that can happen during routine maintenance, with no multi-billion-dollar engine overhauls required. Thus, it makes our SAF very useful and differentiates it from the rest of the SAF.

Keeping Ticket Prices Stable

The biggest obstacle to SAF adoption is cost. If switching to this fuel sends ticket prices through the roof, demand collapses, and the whole transition stalls. To avoid this, we need a multi-pronged economic strategy.

Carbon taxation is shifting the landscape. Under systems like the EU Emissions Trading System (EU ETS) and CORSIA, financial penalties on fossil-derived Jet A-1 are rising fast (Scheelhaase et al., 2018). As conventional kerosene gets more expensive due to these taxes, the relative price gap narrows, making SAF increasingly competitive.

The hybrid biorefinery model also has built-in economic safeguards. Processing municipal waste generates "tipping fees"; municipalities pay biorefineries to handle their waste, which subsidizes capital and operating costs (Wang, 2016). In addition, the HTL and Fischer–Tropsch processes yield valuable byproducts like bio-methanol and nutrient-rich biochar. Selling these to agricultural and chemical markets creates secondary revenue streams that drive down the per-liter fuel cost (Geleynse et al., 2018).

Airlines can capture premium markets too. Corporate clients with strict Scope 3 emission targets will pay extra for certified zero-emission flights. By offering these as premium travel packages, airlines can absorb initial supply chain costs without hiking baseline economy fares.

APPENDICES

Appendix 1. Application to Helicopters

The integration of the novel SAF into the helicopter sector offers a promising pathway for decarbonization, but it presents different technological challenges compared to airplanes. Today, some helicopters already use SAF blends, such as those in the Airbus family (Airbus). Helicopters are powered by turboshaft engines, which fundamentally differ from large commercial turbofans, such as the CFM56-7B evaluated in the original simulation. Turboshafts feature smaller, often reverse-flow combustors, making them highly sensitive to ignition delays and flame stability during the rapid, transient power changes required in real helicopter flight (Chabée and Sauer, 2025).

Historically, the constraint preventing the use of 100% SAF in turboshaft engines is the absence of aromatics, which are required to swell elastomeric seals and prevent fuel leakage, leading to a limitation in commercial helicopter certification at 50% SAF (Chabée and Sauer, 2025). However, the novel hybrid SAF bypasses this integration barrier; its hydrothermal liquefaction stream provides a natural aromatics content of 9.6%, meeting ASTM requirements (Alrebei et al., 2024). Experimental testing on turboshaft engines demonstrates that this SAF improves localized air quality, reducing non-volatile particulate matter (nvPM) number concentrations by up to 81% and emitted particle mass by up to 76%, which is particularly beneficial for urban environment flights (Rabl et al., 2025). Because the novel SAF maintains a combustion temperature almost identical to fossil fuels, it minimizes the changes required to current engines. However, it must be kept in mind that no real-life tests have been conducted (Alrebei et al., 2024).

Despite these strong technological similarities, applying this novel SAF to the helicopter field introduces distinct economic and logistical risks. Helicopter operations are characteristically decentralized, frequently utilizing isolated helipads and offshore platforms rather than centralized international airport hubs. Establishing a fragmented supply chain to distribute the novel SAF to these dispersed bases introduces significant logistical challenges. In addition, integrating advanced conversion technologies within the proposed hybrid biorefinery requires massive initial capital expenditure (CAPEX) (Wang, 2016).

While simulations for this specific novel SAF suggest a potential cost reduction compared to fossil fuels (Alrebei et al., 2024), these margins depend entirely on achieving economies of scale. Given the relatively low total fuel volume consumed by the global helicopter fleet compared to commercial airlines, regional biorefineries may struggle to secure the continuous, high-volume demand required to maintain a low minimum jet-fuel selling price. Consequently, an inability to scale production locally could financially isolate helicopter operators, forcing them to absorb high supply chain costs and delaying the rotorcraft transition to 100% SAF.

Appendix 2. Simulation process and results

On COMSOL software, we decided to study a straight pipeline according to the technical report (a 2 meter long pipe with a 30 mm diameter).

The technical study is carried out under controlled operating conditions to evaluate the behavior of a novel multifeedstock Sustainable Aviation Fuel (SAF) when blended with conventional Jet A-1 kerosene. The SAF considered in this work is produced through a hybrid biorefinery process combining hydrothermal liquefaction, gasification, and Fischer–Tropsch synthesis, resulting in a fuel with approximately 9.6% aromatic content, compatible with ASTM requirements. Four fuel configurations are analyzed: pure Jet A-1, a 95% kerosene and 5% SAF blend, a 50% and 50% blend, and pure SAF. The reference thermodynamic state is set at a temperature of 15°C and a pressure of 1 atmosphere, which corresponds to standard conditions used for fuel property evaluation especially during take-off and landings. These conditions ensure consistent comparison of density and viscosity across all blends, isolating the effect of SAF concentration on fluid behavior.

The flow is then simulated within a pipeline system operating at elevated pressure levels of approximately 50 bars, representative of real aircraft fuel distribution and injection systems. At such pressures, the fluid is subjected to significant mechanical constraints, and even small variations in viscosity or density can lead to measurable changes in hydraulic performance. The geometry used in the COMSOL model includes straight sections and bends, allowing the study to capture both distributed pressure losses (due to viscous friction along the pipe walls) and localized losses (caused by flow separation, secondary flows, and turbulence in curved sections).

These physical properties are then implemented into a COMSOL model representing a simplified but representative section of an aircraft fuel distribution system, typically a pipe with geometric features such as bends that are known to induce additional pressure losses. Boundary conditions are applied in order to reproduce realistic operating conditions, including imposed flow rates or pressures at the inlet, as well as appropriate outlet conditions. The model is done at a steady-state flow regime and focuses on internal fluid dynamics without coupling with thermal or chemical effects, thereby isolating the purely hydraulic behavior of the different fuel blends.

The outputs of the simulation consist of key flow variables that characterize the performance of the system. Among these, pressure distribution along the conduit, velocity fields, and both mass and volumetric flow rates are extracted. These outputs provide a comprehensive picture of how the fluid evolves within the system and allow for the identification of critical zones, particularly in regions such as bends where flow disturbances and localized losses are more important.

The integration of this technical study into the global assessment is therefore crucial. It provides a physical basis for evaluating one of the key technological risks associated with SAF adoption: compatibility with existing infrastructure. If significant deviations in flow behavior were observed, this would imply the need for system modifications, such as increased pumping power or redesign of certain components, which would directly impact both capital and operational expenditures. Conversely, if the variations remain moderate, the fuel can be considered compatible with minimal adaptation, reinforcing its economic and operational viability.

In this context, the COMSOL study is a bridge between theoretical fuel properties and real-world implementation constraints. As such, it plays a central role in supporting the techno-economic analysis by quantifying potential integration issues and informing decision-making regarding the feasibility of large-scale SAF deployment in the aviation sector.

Here are different figures illustrating our simulation.

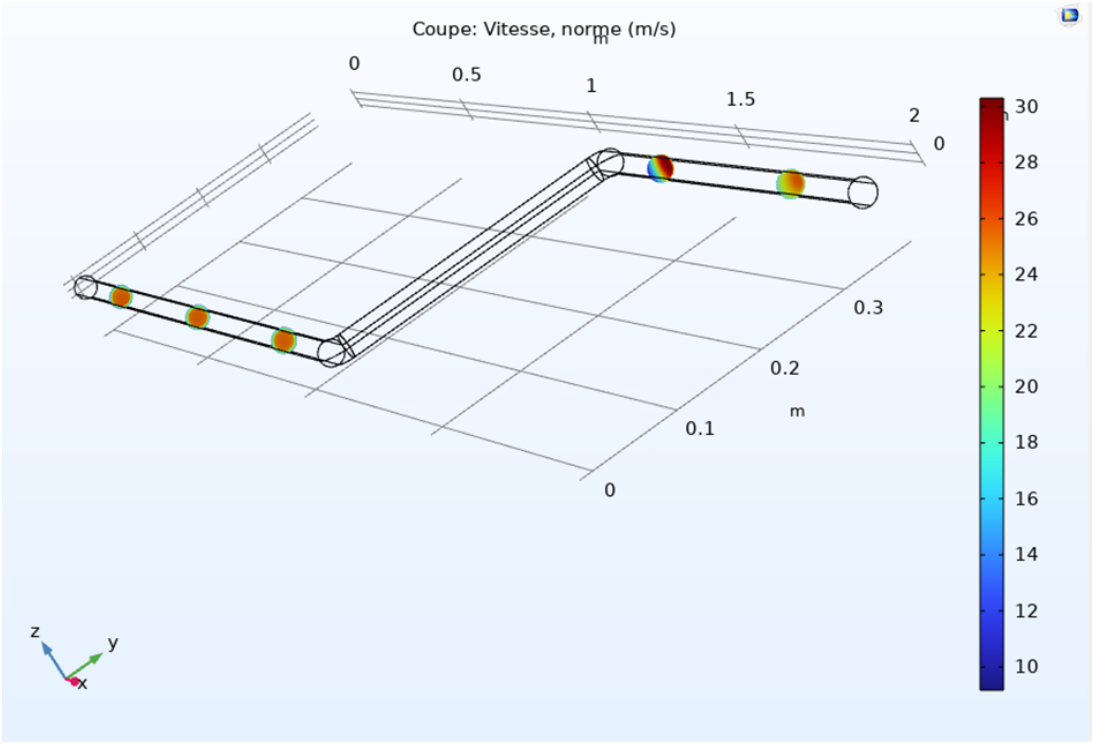


Fig. 1: Pipe velocity cross-section diagram

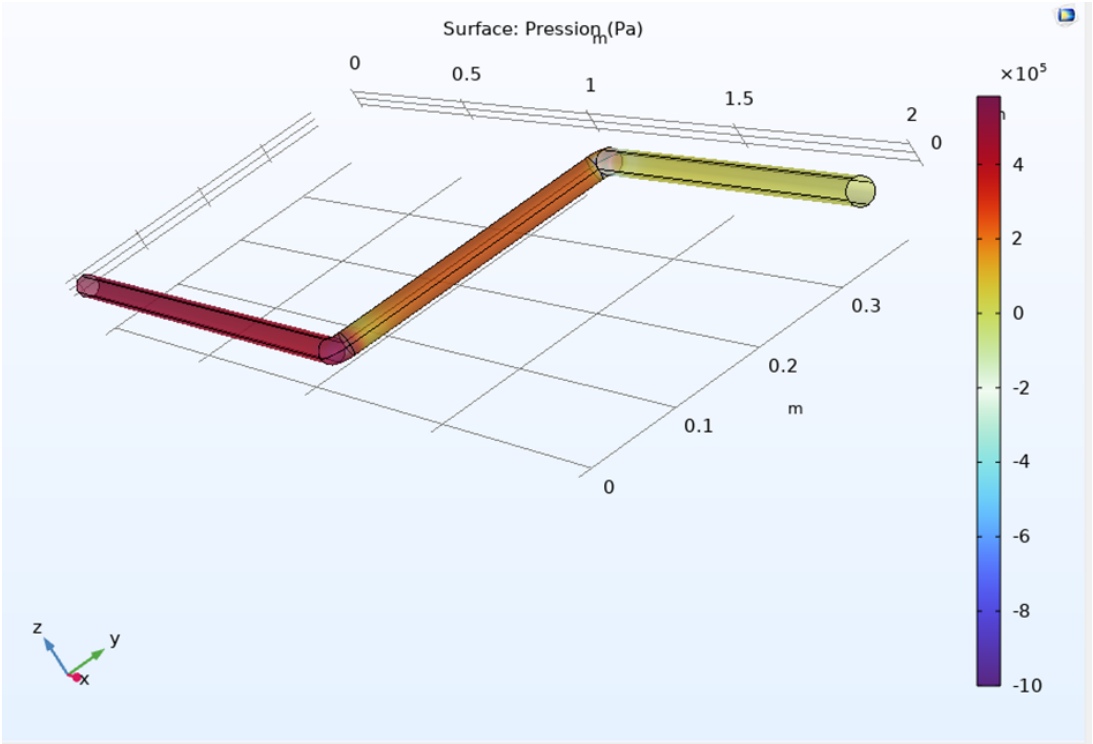


Fig. 2: Area and pressure diagram along the pipe

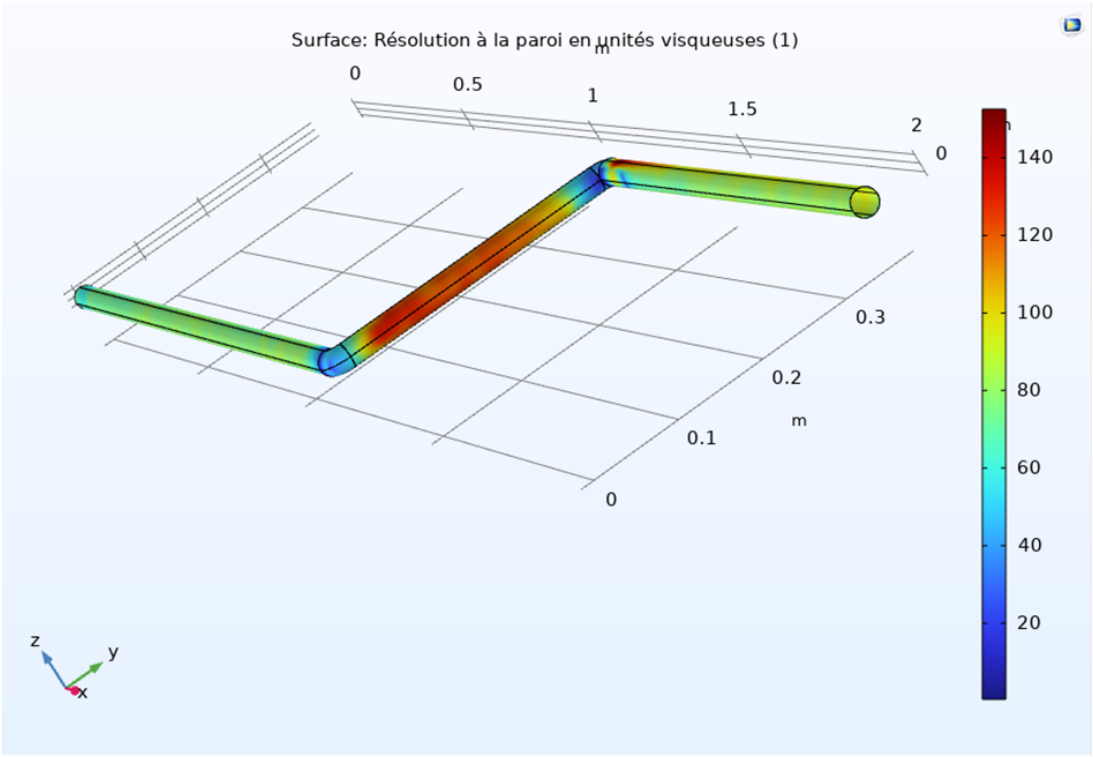


Fig. 3: Surface: Wall resolution in viscous units

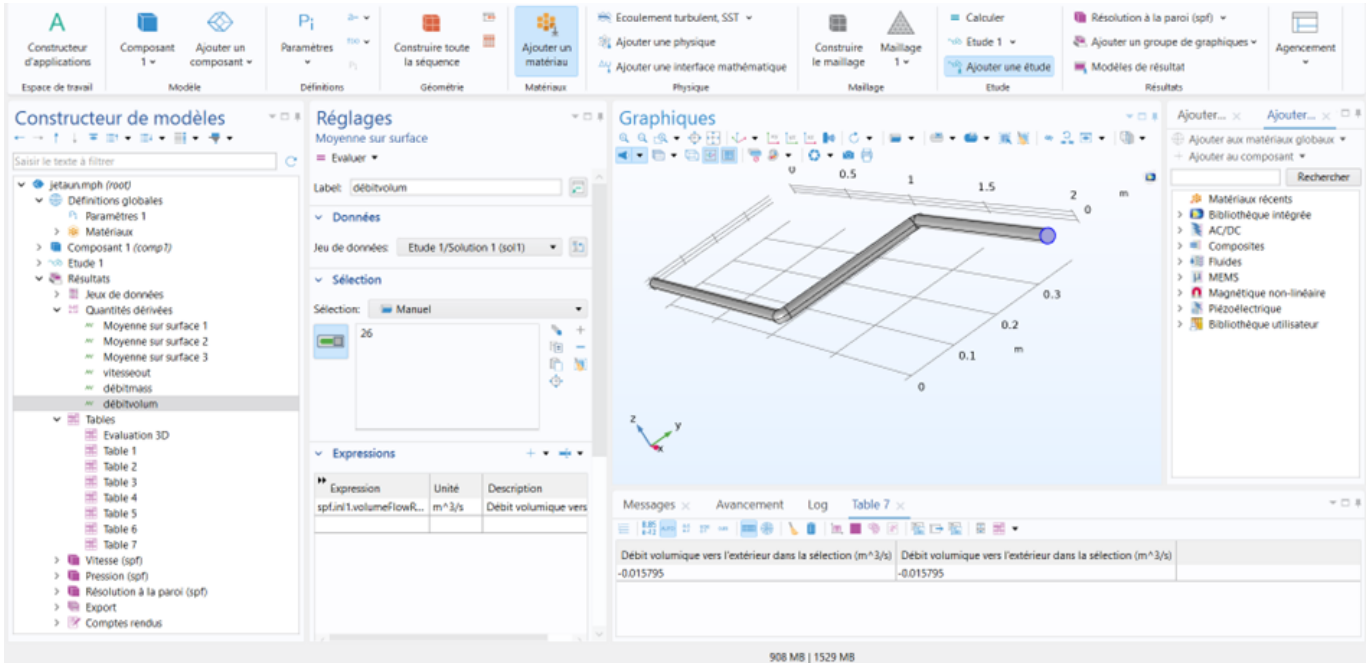


Fig. 4: Mass flow rate outward through the selection outlet (kg/s)

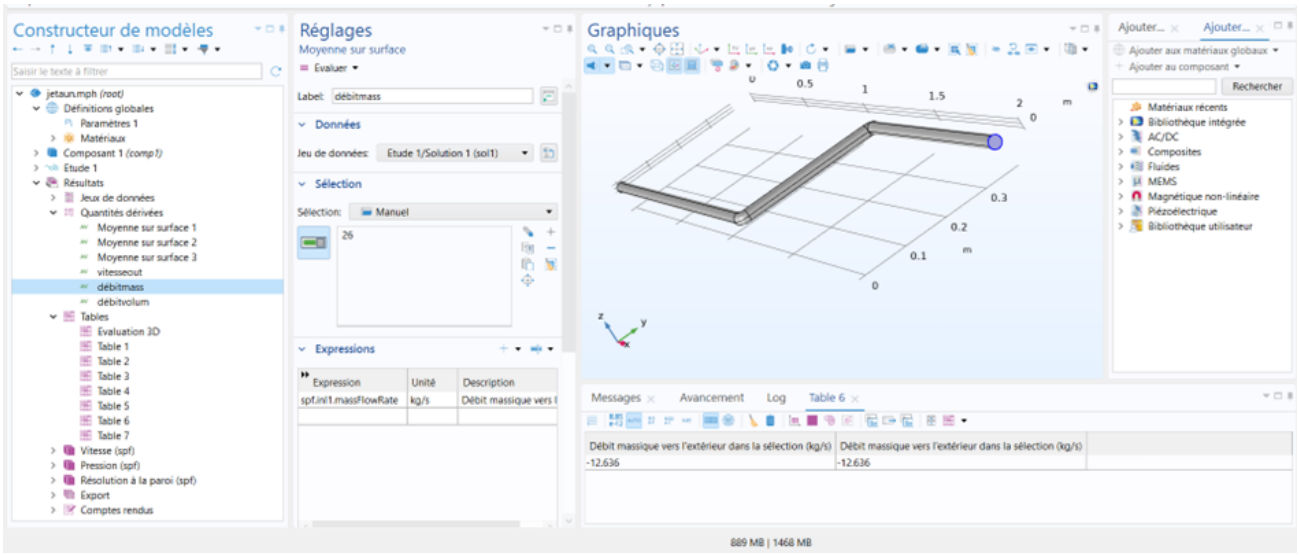


Fig. 5: Mass flow rate outward through the selection outlet $(\mathrm{kg/s})$

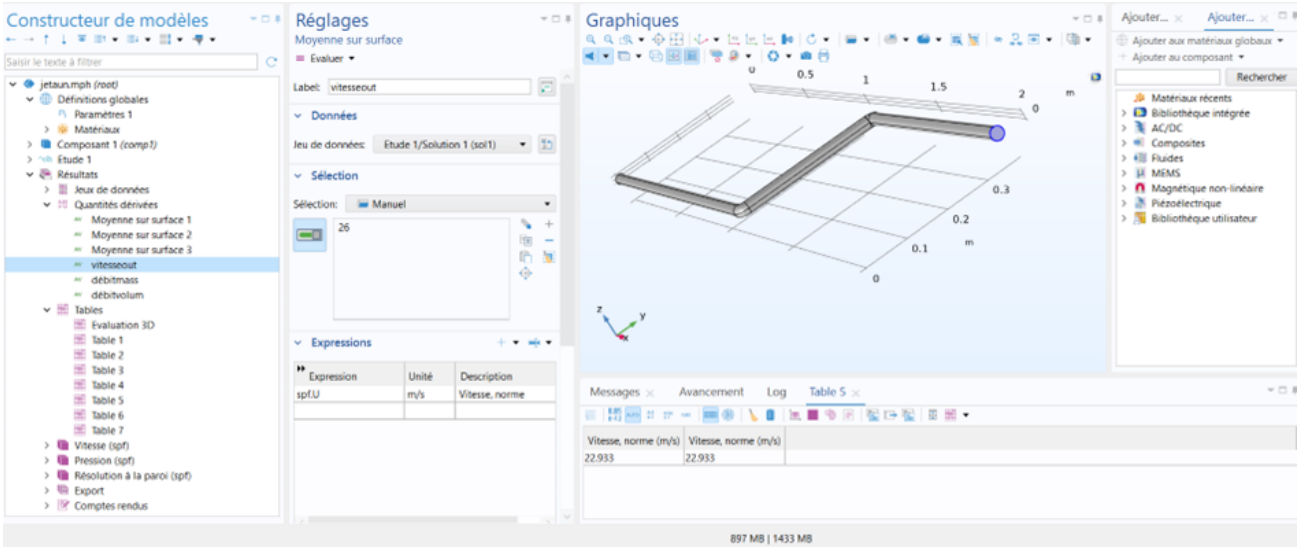


Fig. 6: Screenshot of the geometry and velocity of the inlet/outlet

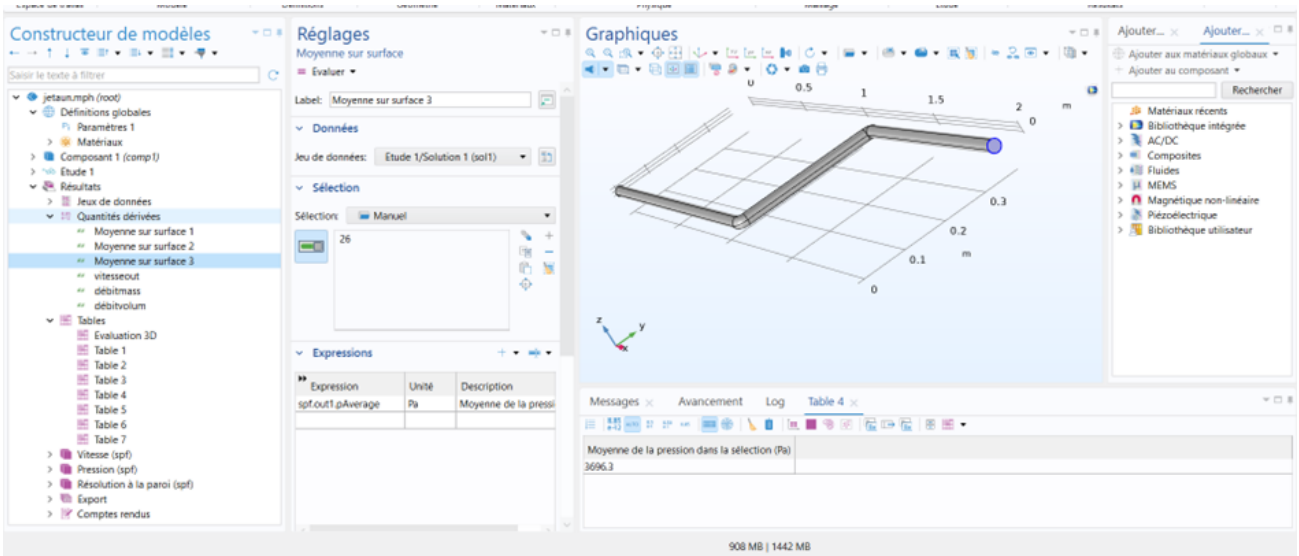


Fig. 7: Screenshot of the geometry used in the simulation with pressure

The velocity histograms show a distinct shift in distribution toward lower values as the SAF concentration increases. Specifically, the peak for pure SAF is positioned at a lower velocity than that of pure Kerosene, confirming that the synthetic fuel yields a lower average exit velocity of 21.487 m/s compared to 22.933 m/s for the baseline. This shift indicates that the SAF siphons off kinetic energy more rapidly, resulting in a less "energetic" flow within the aircraft's internal piping.

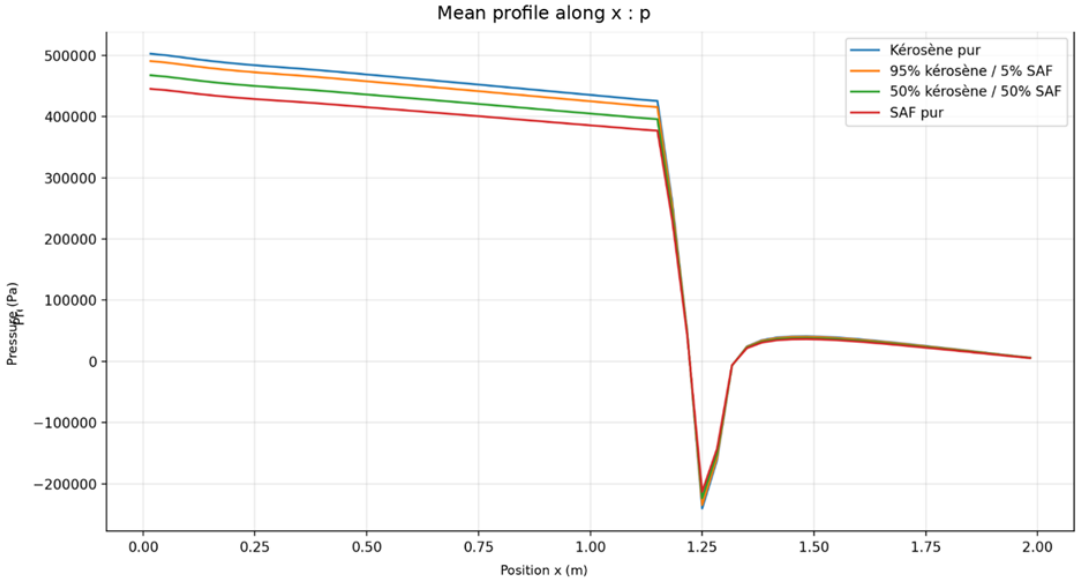


Fig. 8: Mean profile along x : P



Fig. 9: Mean profile along x : spf.U

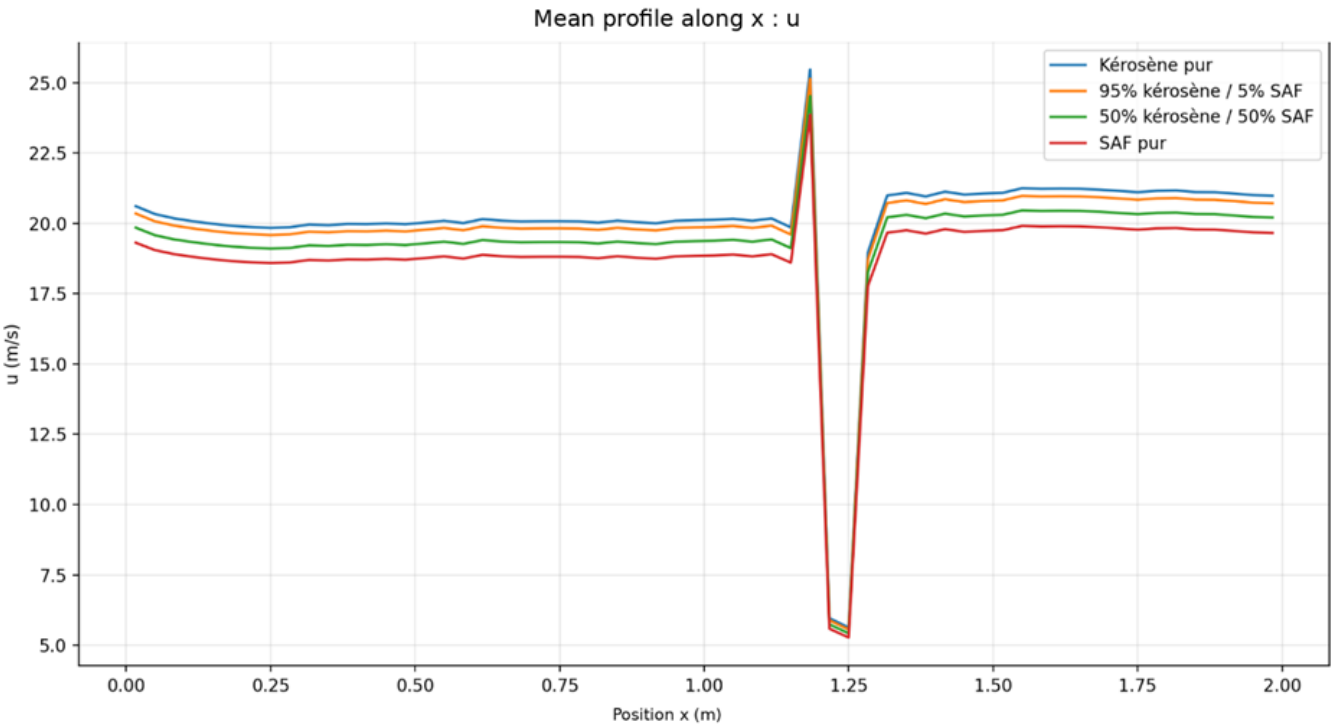


Fig. 10: Mean profile along x : u

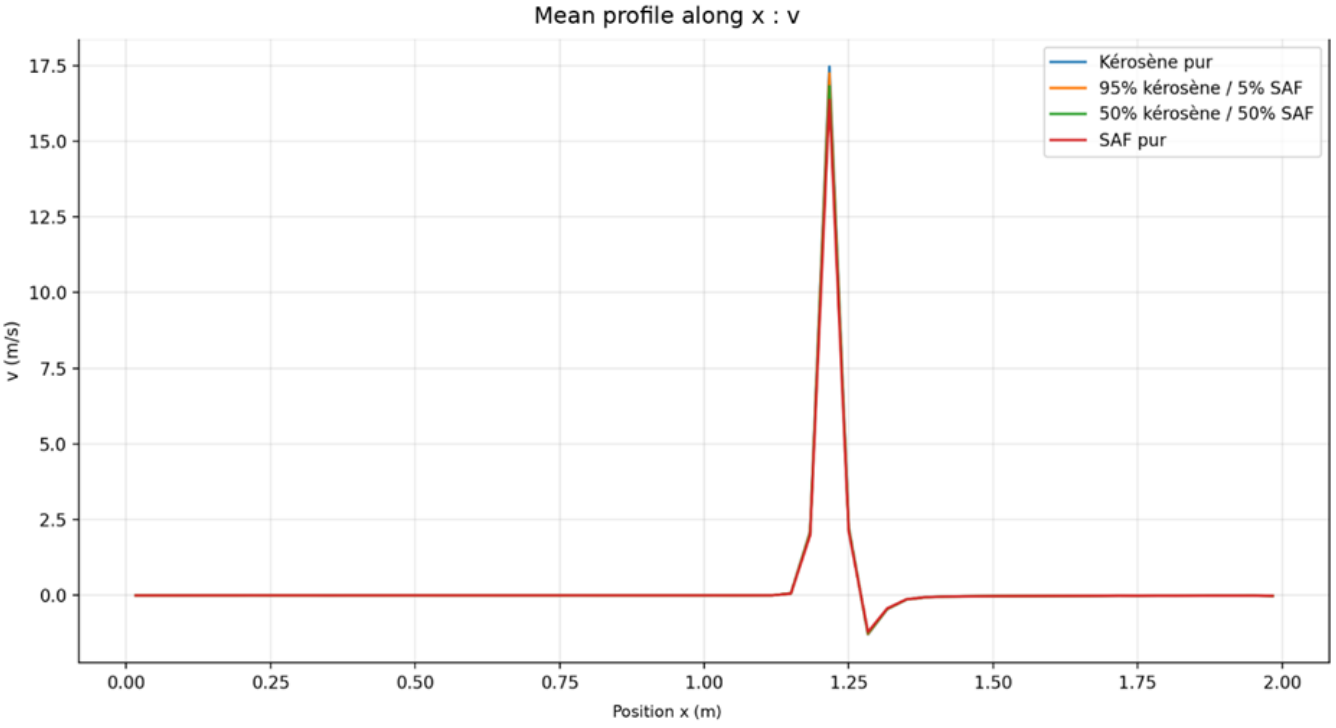


Fig. 11: Mean profile along x : v

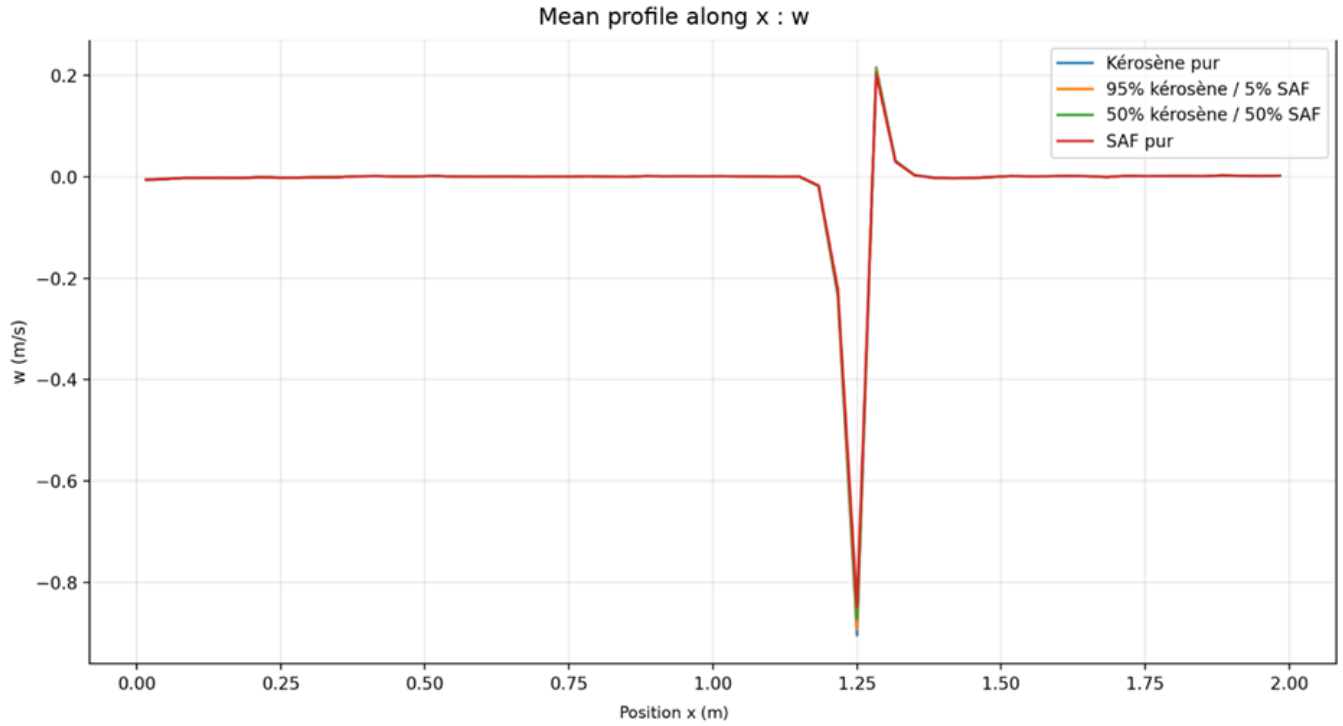


Fig. 12: Mean profile along x : w

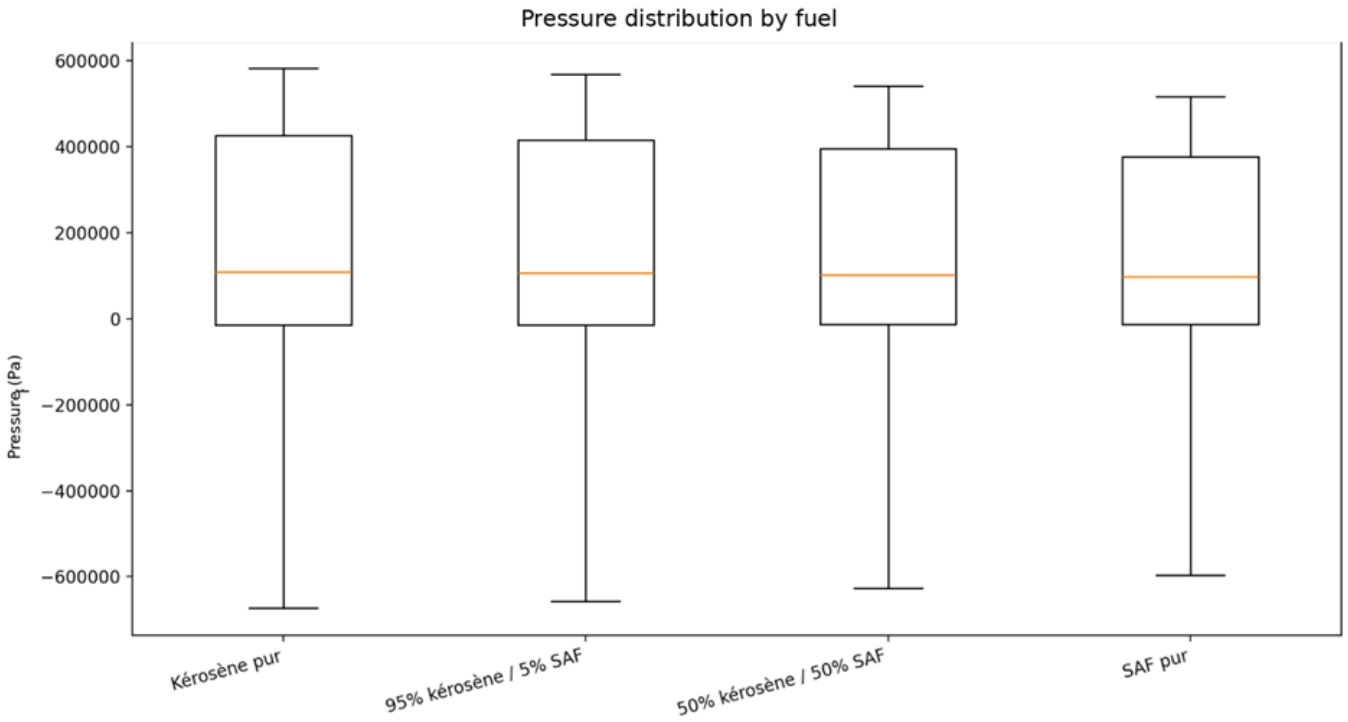


Fig. 13: Pressure distribution by fuel

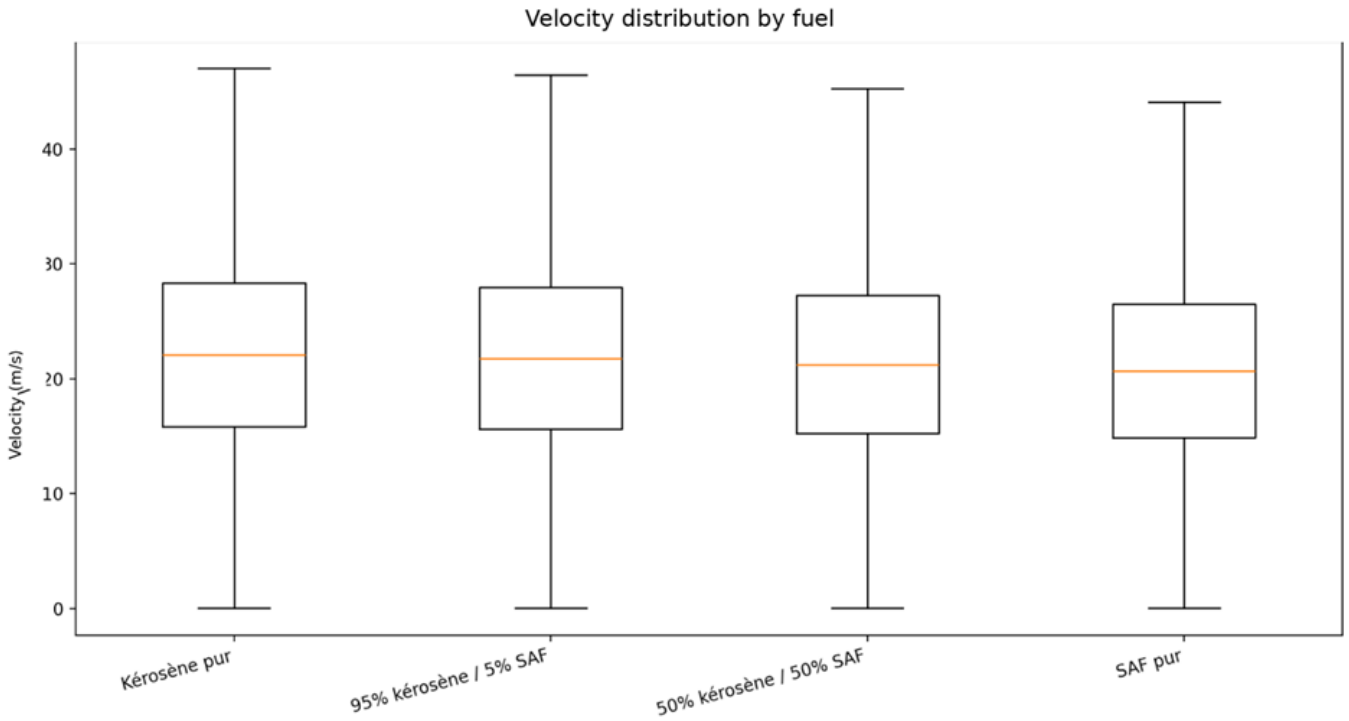


Fig. 14: Velocity distribution by fuel

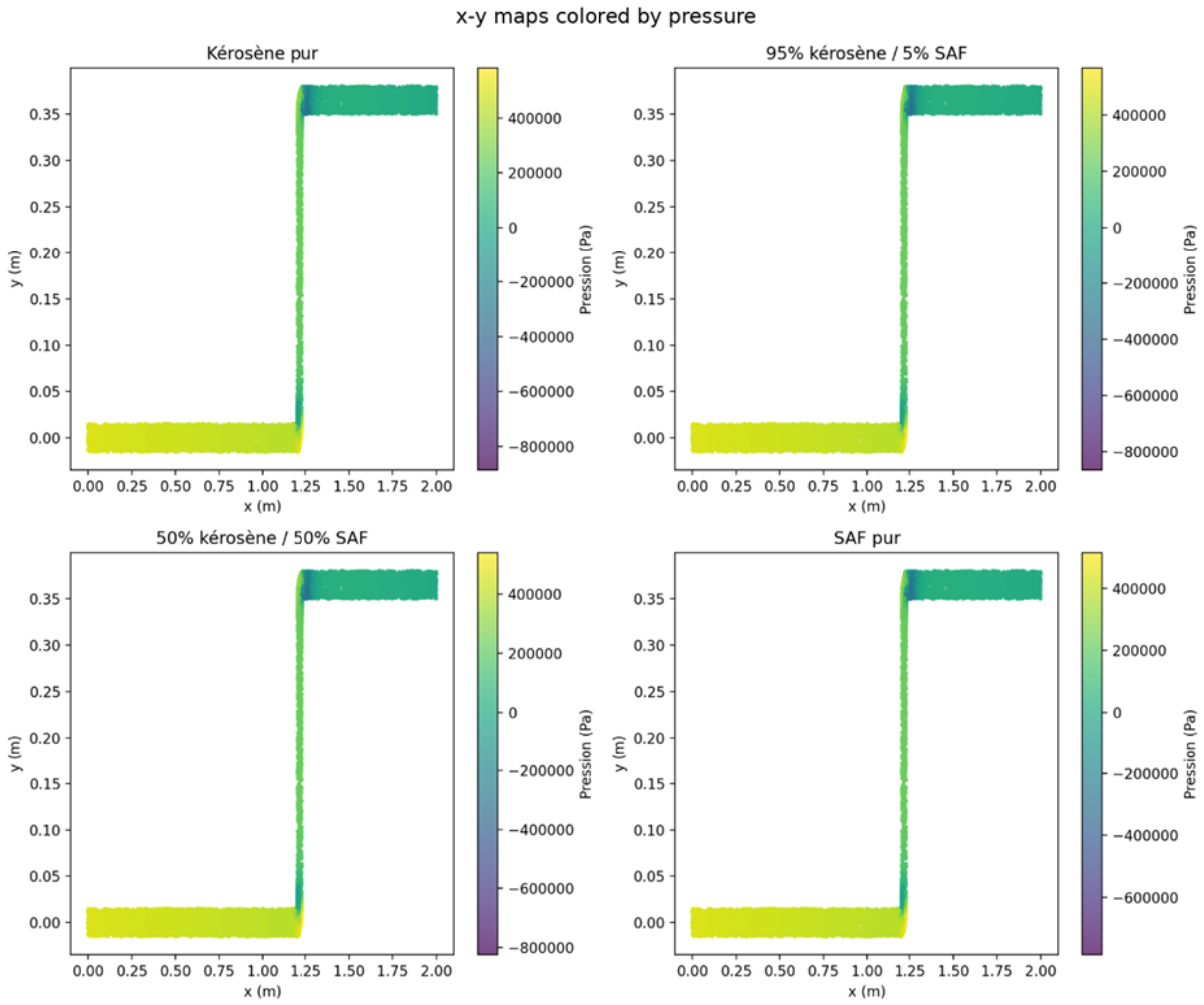


Fig. 15: X-Y maps colored by pressure diagram

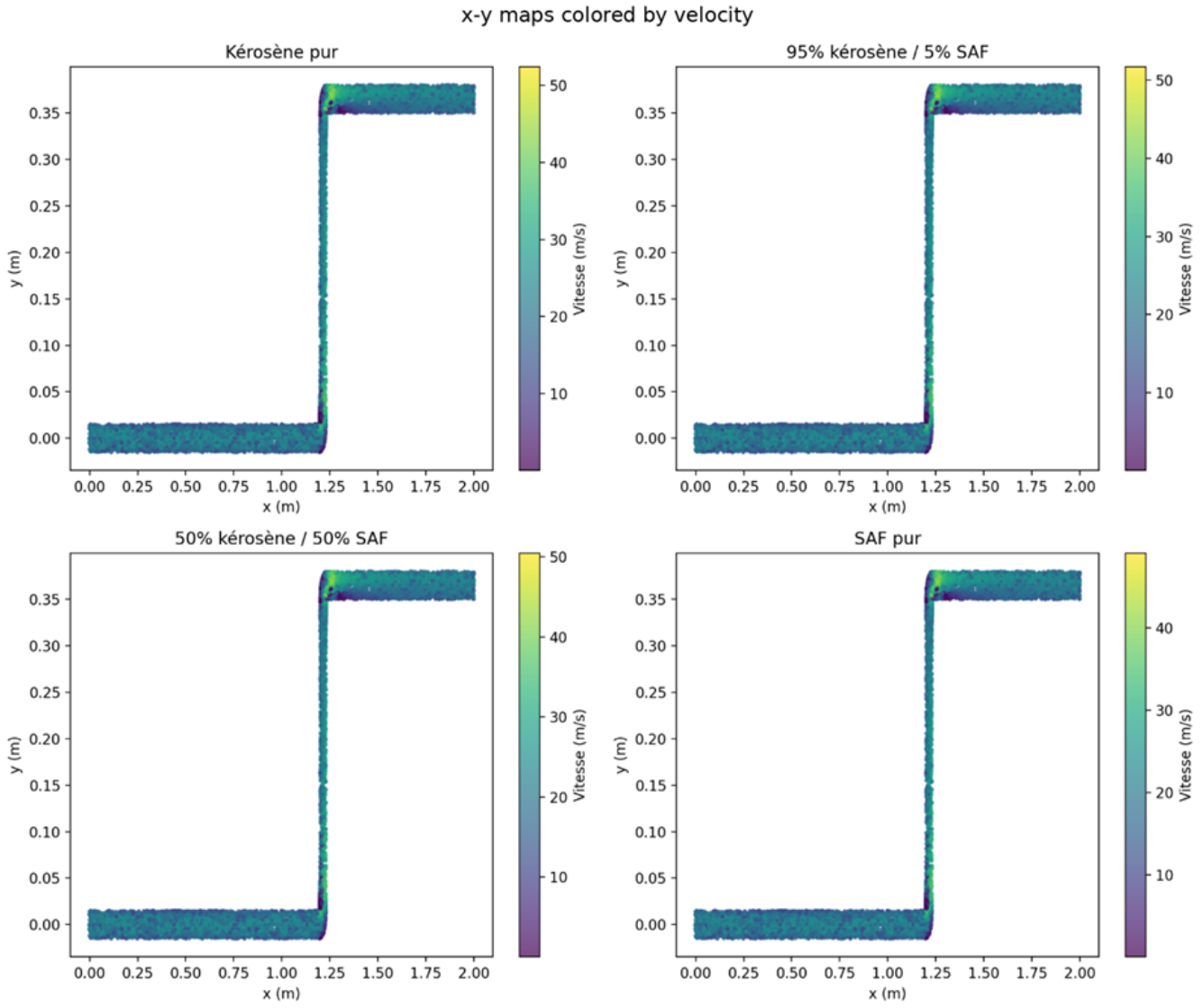


Fig. 16: X-Y maps colored by velocity diagram

The spatial x-y maps for pressure and velocity further reveal how the 90-degree pipe bend affects the fluid. In the pressure maps, the high-pressure zones (yellow) at the inlet are visibly reduced for the SAF scenarios, and a sharper pressure drop occurs immediately following the elbow. The velocity maps highlight darker areas in the inner corner of the bend, signifying flow separation and stagnation points. Because the SAF has a lower density and different viscosity profile,

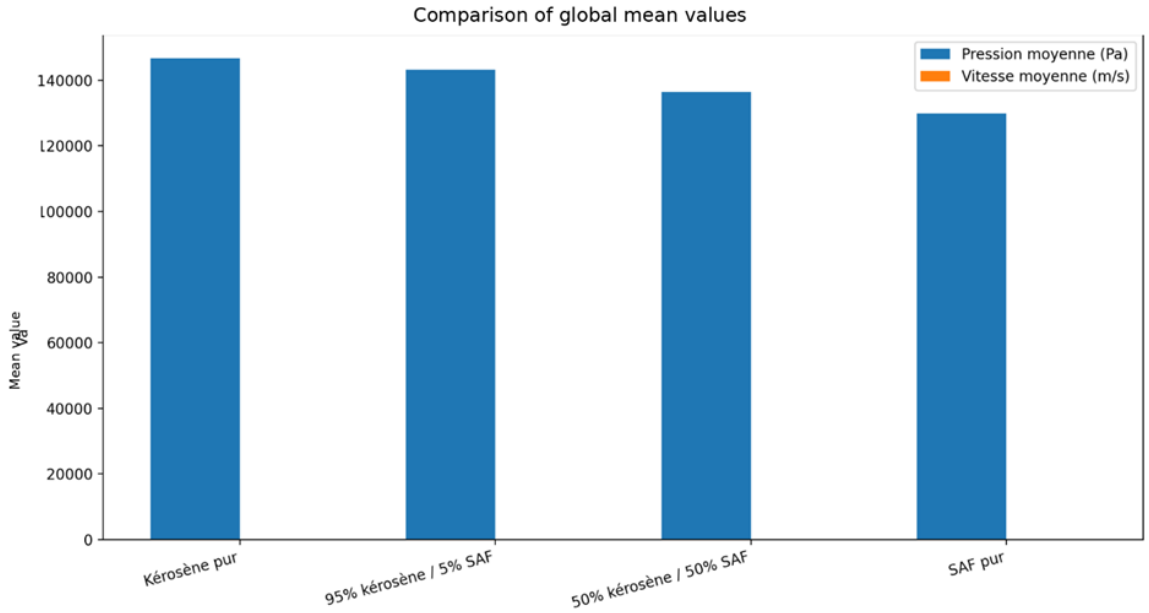


Fig. 17: Comparison global mean values of all kind of fuels

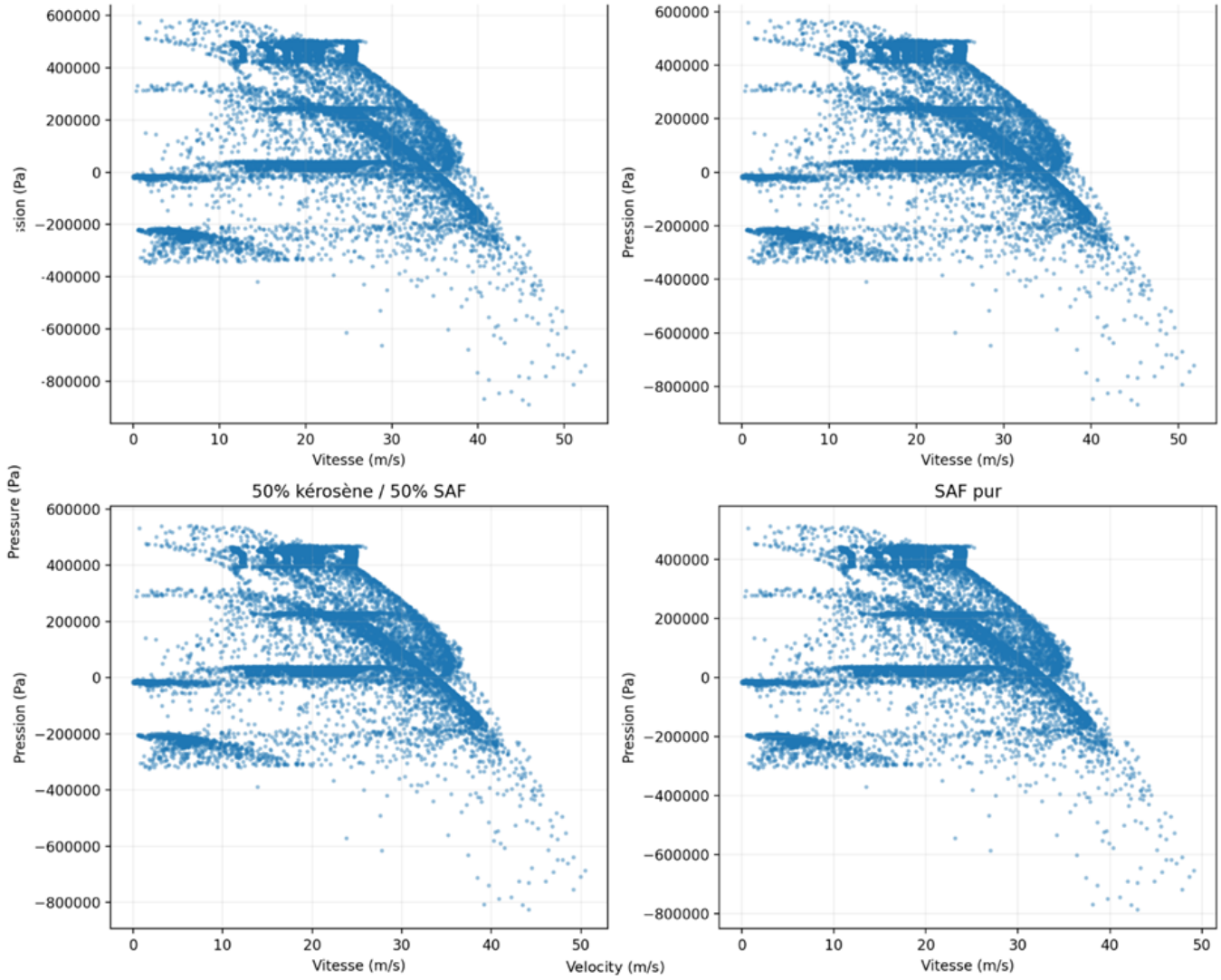


Fig. 18: Velocity diagram with all different fuels

The pressure-velocity correlation plots demonstrate that while the SAF is a functional “drop-in” replacement, it is hydraulically less robust. The scatter plots show that the overall flow behavior remains predictable and consistent across all blends, which supports the fuel’s viability for existing aircraft. However, the entire data cloud for the pure SAF sits lower on the pressure axis.

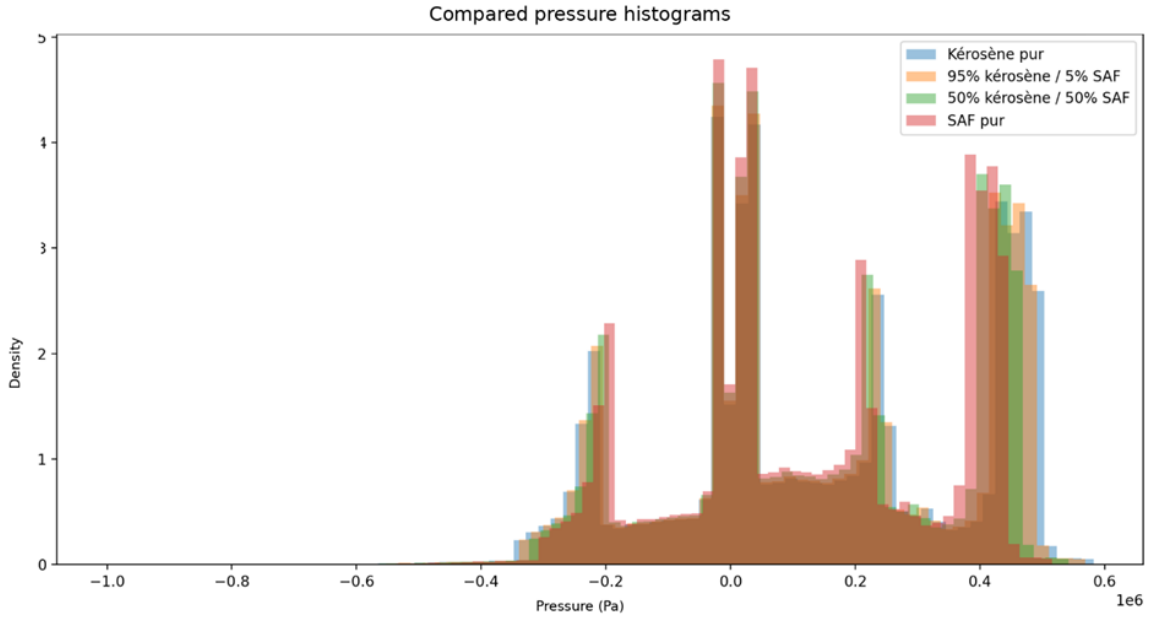


Fig. 19: Compared pressure histograms with different configurations

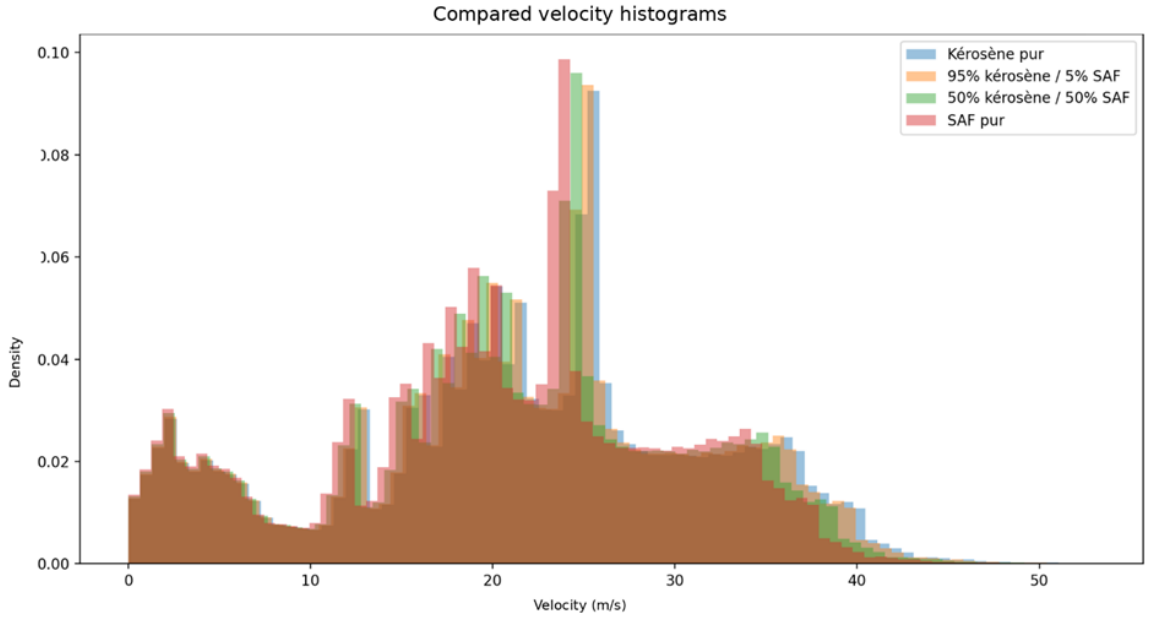


Fig. 20: Compared velocity histograms with different fuel configurations

The velocity histograms show a distinct shift in distribution toward lower values as the SAF concentration increases. Specifically, the peak for pure SAF is positioned at a lower velocity than that of pure Kerosene, confirming that the synthetic fuel yields a lower average exit velocity of 21.487 m/s compared to 22.933 m/s for the baseline. This shift indicates that the SAF siphons off kinetic energy more rapidly, resulting in a less “energetic” flow within the aircraft’s internal piping.

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